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AI Disclosure in Annual Reports: Evolution, Determinants, and Capital Market Participants' Perceptions

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ABSTRACT: We provide large-sample evidence on the prevalence, determinants, and capital market consequences of artificial intelligence (AI) disclosures in U.S. firms' Form 10-K filings from 2017 to 2024. Using comprehensive AI-related keywords and a large language model, we document a gradual rise in AI disclosure prior to ChatGPT and sharp increases thereafter, particularly in Risk Factors. Post-ChatGPT, AI risk disclosures become longer, more specific, and less boilerplate. Firms' disclosure of AI-related information is strongly associated with *ex ante* exposure to AI technologies and *ex post* AI-related negative incidents. Examining market reactions, we find no association between these disclosures and information environment quality prior to ChatGPT's release. After ChatGPT, AI disclosure among high-exposure firms is associated with lower post-filing volatility but higher analyst forecast dispersion and error, suggesting divergent interpretations by investors and analysts. Our findings inform ongoing debates on AI transparency and disclosure regulation.

Data Availability: Data used in this study are publicly available.

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Keywords: AI disclosure; AI risk; AI ethics, AI opportunity; large language models; textual analysis; market uncertainty.

I. INTRODUCTION

The past decade has seen dramatic advancements in artificial intelligence (AI). As a general-purpose technology, AI is increasingly embedded across industries, reshaping firms' operations and strategies (Agrawal, Gans, and Goldfarb 2018; Stanford University Human-Centered Artificial Intelligence (Stanford HAI) 2024). Although AI offers substantial potential for innovation and efficiency (Babina, Fedyk, He, and Hodson 2024), it also raises profound ethical challenges related to bias, discrimination, misinformation, data privacy, and data security—issues that sit

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All errors are solely ours.

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Supplemental material is available online, as linked in the text.

Any use of generative artificial intelligence (AI) or AI tools by the authors has been disclosed in the "Declaration of Generative AI and AI-Assisted Technologies" section.

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at the core of responsible AI governance (United Nations Educational and Scientific and Cultural Organization (UNESCO) 2021; U.S. Senate Committee 2023).

As investors increasingly incorporate AI-related risks and opportunities into decision-making (Bloomberg 2023), AI disclosures in corporate filings have grown. Regulators have expressed concern about the accuracy and substance of these disclosures. The former chair of the U.S. Securities and Exchange Commission (SEC) warned firms against misleading AI-related statements (Gensler 2023), and in 2024, the SEC brought enforcement action against an AI recruitment startup for "AI washing," alleging the use of AI-related buzzwords to mislead investors (SEC 2024b).[1] At the same time, activist shareholders are demanding greater transparency around firms' AI usage, governance, and exposure to risks related to human rights, misinformation, and disinformation (Hearon, Cave, and O'Brien 2024).

Despite the growing importance of AI disclosure, little is known about its current state. How prevalent are AI-related disclosures in corporate filings? Where do they appear, what do firms discuss, and which risks are emphasized? Which firms disclose AI, and how do capital market participants respond? Our study addresses these questions by compiling large-sample evidence on the prevalence, characteristics, determinants, and market perception of AI disclosures in annual SEC Form 10-K filings. Understanding the existing disclosure landscape is a necessary first step for informed discussion of disclosure mandates (Leuz 2018).

Our first objective is to document trends and characteristics of AI disclosure in Form 10-K filings from 2017 to 2024, a period that coincides with the commercialization of AI (McKinsey 2022, 2024). Using a comprehensive AI keyword list from prior literature (Chen and Srinivasan 2024; Anantharaman, Rozario, and Zhang 2024), we find that 11.25 percent of filings mention AI in the Item 1 Business Description, 7.78 percent in Item 1A Risk Factors, and 4.24 percent in Item 7 MD&A. Even in 2022 (the pre-ChatGPT era), 15 percent of firms discussed AI in the Business Description. Although technology firms disclose most frequently, manufacturing and consumer durables firms also exhibit substantial AI discussion.

AI mentions increase gradually prior to 2023 and rise sharply following the release of ChatGPT, particularly in Risk Factors. The number of firms first mentioning AI in Risk Factors increases nearly eightfold from 2022 to 2024, with distinct spikes following the March 2023 open letter calling for a pause in AI development and the October 2023 White House AI executive order.[2]

We analyze six characteristics of AI disclosure: length, sentiment, year-on-year dissimilarity, specificity, topical content, and whether AI use is planned or realized, leveraging GPT-4o. AI-related discussions—especially risk disclosures—have become longer over time. Sentiment remains stable in Business Descriptions, grows more negative in Risk Factors, and becomes increasingly positive in MD&A, reflecting the dual framing of AI as both opportunity and risk. Beginning in 2023, AI disclosures in Risk Factors have become more specific and less similar to prior-year disclosures, consistent with SEC calls for tailored, nonboilerplate risk disclosure. In contrast, Business Description and MD&A discussions show little change in specificity, echoing regulatory concerns about generic AI disclosures.

Topic modeling reveals that Business Description and MD&A discussions focus primarily on AI's role in enhancing products and services, improving operational efficiency, accelerating R&D, supporting marketing and customer engagement, and improving decision-making. Risk Factor disclosures emphasize security and privacy, reputational harm, regulatory and legal risk, ethical and social concerns, and financial risk. Most AI discussions describe existing AI adoption (73 percent in Business Description; 62 percent in MD&A), although both intended and realized adoption rise sharply after ChatGPT's release, suggesting that generative AI accelerated firms' AI deployment.

Our second objective is to examine the determinants of AI disclosure. Larger firms and firms with greater analyst following are more likely to disclose AI, whereas firms with higher institutional ownership are less likely to discuss AI-related risks. Firm fixed effects add considerable explanatory power, indicating firm-level, time-invariant drivers of AI disclosure. Disclosure persistence is strong once firms begin discussing AI.

We construct two measures of firms' exposure to AI: an *ex ante* measure based on the similarity of each firm's Business Description to AI-related patents filed till date and an *ex post* measure of AI-related negative incidents experienced by the firm.[3] Both measures are strongly associated with AI disclosure. AI-related negative incidents are particularly associated with firm's AI risk disclosure (especially in the following year), suggesting that problematic experiences

[1] In their list of disclosure priorities for 2024, the SEC describes that it will consider how firms disclose AI-related opportunities and risks in annual reports, including how a firm defines AI, the level of customization of the disclosures, and whether the firm has a reasonable basis for making claims about the prospects of AI. See Gerding (2024) and SEC (2024a).

[2] The letter argued that AI development posed "profound risks to society and humanity" and was signed by tech industry leaders. See Future of Life Institute (2023).

[3] For example, the Federal Trade Commission accused Rite Aid of failing to take reasonable measures to prevent harm to customers when using AI-based facial recognition technology from 2012 to 2020 to identify those they suspected of shoplifting or other problematic behavior. As a result, Rite Aid was banned from using AI facial recognition for five years. See Federal Trade Commission (2023).

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with AI—often related to issues of algorithmic bias—prompt reactive transparency. However, firms in highly litigious industries are less forthcoming after such incidents. AI exposure also affects the content of disclosure—firms with high *ex ante* AI exposure provide more balanced discussions of risks and opportunities, whereas low-exposure firms emphasize risks.

Our third objective is to study market reactions to AI disclosure before and after ChatGPT, which increased public awareness of AI dramatically. Using data from 2021 to 2023, we find no association between AI disclosure and the quality of the firm's information environment (i.e., post-filing volatility, analysts' forecast error and dispersion) prior to ChatGPT. Post-ChatGPT, however, AI disclosure among high-exposure firms is associated with lower post-filing volatility but higher analysts' forecast error and dispersion. These effects are stronger for longer, more positive, more specific, and more wide-ranging discussions of AI. The divergent responses suggest that investors perceive AI disclosure as reducing uncertainty, whereas analysts—arguably more sophisticated users of financial information—interpret the same disclosures as increasing uncertainty.

Our study contributes to the accounting literature on voluntary disclosures in SEC filings as well as a rapidly growing, interdisciplinary literature that aims to understand the impact of AI technologies on business. It complements evidence from Jia, Li, Ma, and Xu (2024), who examine AI mentions in conference calls. Although conference calls are a less regulated channel that allows more candid discussion, annual reports offer the most standardized disclosure avenue available across the broad universe of listed firms. Differences between the channels are reflected in the findings: Jia et al. (2024) focus on generative AI mentions post-ChatGPT and document predominantly positive sentiment and managerial optimism; in contrast, we study AI disclosures more broadly across the full 2017–2024 period and document a gradual rise in annual report mentions of AI even over the pre-ChatGPT era, finding a more nuanced picture with negative sentiment increasing in Risk Factors even as MD&A discussions grow more positive. Where Jia et al. (2024) document a dissonance between managers' and investors' perceptions of AI, we uncover a dissonance between investors' and analysts' perceptions of AI disclosure post-ChatGPT.

Our study also adds to the information systems (IS) literature on AI governance and transparency. By documenting how and when firms disclose AI-related risks and opportunities in corporate annual reports, we speak directly to discussions about corporate responsibility in AI deployment, a key concern in IS research on responsible AI (Lyytinen, Nickerson, and King 2021). Lastly, our findings on divergent investor and analyst reactions to AI disclosure contribute to discussions on the role of human-AI interaction in accounting and financial information processing by showing that stakeholders differ in how they interpret AI-related information in annual reports.

II. BACKGROUND

Artificial intelligence (AI) is a multidisciplinary field aimed at developing intelligent machines, distinct from the natural intelligence exhibited by humans and animals (Hogarth and Benaich 2021; Russell and Norvig 2016). Extant research highlights the benefits of AI across various firm outcomes. Chen and Srinivasan (2024) demonstrate that investments in digital technologies, including AI, are linked to higher market valuations. Using a resume-based measure of AI-related human capital investment, Babina et al. (2024) document a positive relationship with product innovation and sales growth. AI adoption has been shown to enhance the quality of financial reporting (Anantharaman et al. 2024) and earnings forecasts (Brown, Louis, Rozario, and Zhang 2024) while decreasing the likelihood of internal control weaknesses (Ashraf 2025).

Despite the numerous benefits of AI, firms face significant costs and risks associated with its adoption that go beyond direct investment costs (Nayak and Tiyadi 2024). Foremost are challenges arising from AI-induced disruptions to the fundamental operating model of many sectors. AI technologies' capabilities in automating repetitive tasks, gathering information, and improving decision-making can automate entire functions and render traditional business models obsolete, with effects already evident in knowledge-focused sectors like the media, education, healthcare, legal services, and portfolio management, to name a few.[4]

In addition, firms must contend with litigation risks, reputational costs, and financial reporting issues arising from AI usage. Concerns extend to AI's impact on individuals, including biases, discrimination, misinformation, and data privacy and security risks (U.S. Senate Committee 2023). For instance, AI-induced biases can perpetuate unconscious discrimination in hiring and promotion decisions, increasing firms' legal liability. Similar risks arise from misuse of personally identifiable information and reputational fallout from algorithmic bias incidents (Hirsh and Cha 2018;

[4] See, for example, Johnson (2023) and MIT Technology Review Insights (2024). In early 2026, the software sector faced a particularly intense reckoning and re-evaluation from investors and analysts, amidst rising concerns that new agentic AI tools for software development could replace legacy software products and/or render them obsolete or cost-ineffective for enterprise clients.

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Romanosky, Hoffman, and Acquisti 2014; Perols and Murthy 2021; Holweg, Younger, and Wen 2022). Gary Gensler, former head of the SEC, has even warned of system-wide risks from AI, arising from the widespread reliance on a small number of AI models that could heighten systemic vulnerabilities and potentially trigger financial crises (Sorkin, Mattu, Kessler, De La Merced, and Livni 2023).

Motivated by the increasing use of AI across various business functions and its significant benefits counterweighted by equally significant risks, we systematically investigate AI-related disclosure in firms' annual reports. Our first research question is:

RQ1: What do firms disclose about AI in 10-K reports over time?

Given that AI-related disclosure has voluntary elements (within the constraints of various SEC rules requiring the disclosure of financially material information), we next investigate the determinants of firm disclosure on AI. This is our second research question:

RQ2: What are the determinants of AI disclosure in 10-K reports?

Finally, as the decision to disclose AI information reflects managers' cost-benefit assessments, understanding its implications for capital markets is essential. Prior research suggests that voluntary disclosure can have potentially countervailing effects on information asymmetry across market participants. On one hand, more disclosure can resolve uncertainty by providing useful information for decision-making. On the other hand, it can increase uncertainty by introducing new, complex, or difficult to interpret information (e.g., Lang and Lundholm 1996; Kravet and Muslu 2013). In the context of AI, disclosures may reduce information asymmetry by clarifying firms' strategic positioning and exposure to emerging technologies. However, the very nature of AI makes the counterargument especially acute—these technologies are complex, hard to penetrate, and rapidly evolving, and so AI disclosures may also increase uncertainty and disagreement among market participants. These opposing forces motivate our third research question examining how capital market participants respond to AI disclosure:

RQ3: What are the capital market participants' perceptions of AI disclosure?

III. RESEARCH DESIGN

To identify AI discussions and the locations of those discussions within 10-K filings, we rely on a list of AI keywords established in prior studies (Chen and Srinivasan 2024; Anantharaman et al. 2024).[5] In addition to identifying whether and in which sections firms discuss AI, we extract sentences containing AI-related keywords (hereafter referred to as "AI discussions").

Evolution of AI Disclosure

We begin by examining the evolution of AI disclosure from seven perspectives: occurrence, length, sentiment, year-on-year dissimilarity, specificity, topics, and planned use of AI versus actual use of AI. First, we identify the sections within the 10-K where firms discuss AI and the distribution of AI disclosure over time and by industry. Then, using the extracted sentences containing AI keywords, we measure the length of AI discussions based on word count. We also assess the relative length of these discussions by calculating the proportion of their word count relative to the total word count of the section in which they appear. We analyze the sentiment of AI discussions, with sentiment scores ranging from −1 to 1, where 0 indicates neutrality and positive (negative) values reflect positive (negative) sentiment.[6]

For firms that mention AI consistently over time, it is important to assess whether there are differences in the discussions over time. To examine this, we measure the amount of additional information by calculating the Euclidean distance between the embeddings of AI discussions from year t and year $t-1$ for each firm. Embedding represents text content as a high-dimensional vector that captures its semantic and contextual meaning. We use OpenAI's

[5] These keywords are "artificial intelligence," "AI," "machine learning," "deep learning," "neural network," "supervised learning," "unsupervised learning," "reinforcement learning," "natural language processing," "natural language generation," "NLP," "speech recognition," "facial recognition," "computer vision," "image processing," "image recognition," "video processing," "robotic process automation," "intelligent automation," and "robotics." We add keywords of generative AI technologies, including "generative AI," "generative artificial intelligence," and "large language model."

[6] Our sentiment scores are computed using TextBlob (https://textblob.readthedocs.io/en/dev/quickstart.html), an open-source Python package that applies a lexicon-based (i.e., dictionary-based) sentiment analysis approach. Specifically, TextBlob relies on the Pattern (https://github.com/clips/pattern?tab=readme-ov-file) sentiment lexicon, in which words are manually assigned polarity scores. For each AI-related disclosure, TextBlob aggregates word-level polarity scores to generate a document-level sentiment score.

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"text-embedding-ada-002" model, renowned for its state-of-the-art performance in encoding semantic meaning across diverse text inputs. A larger distance between the embeddings of AI discussions from year t and year $t-1$ for each firm indicates more different information in the current disclosure compared with the previous year. In addition to assessing differences over time (time-series), we also examine differences in the cross-section. This is measured as the Euclidean distance between the embedding of a firm's AI-related discussion in year t and the average embedding of AI-related discussions from other firms within the same two-digit SIC in the same year. A larger value indicates different information disclosed by this firm compared with industry peers in the same year.

The level of specificity in AI-related discussions can also vary across firms and over time. We measure the specificity of these discussions using OpenAI's GPT-4o model. We prompt GPT-4o to read each AI-related discussion and assign a specificity score from 1 to 5, where 1 indicates very general discussions, characterized by vague mentions of AI without much detail, and 5 represents highly specific discussions, offering in-depth, comprehensive insights into AI, including detailed examples and contexts.

To systematically analyze the content of firms' AI-related discussions, we further utilize GPT-4o to identify the main topics covered in these discussions and to label whether a firm has addressed specific topics. For AI-related discussions within each section, we follow these steps: (1) *Topic Identification*: we aggregate all AI-related discussions across firms and years and use the GPT-4o application programming interface (API) to identify distinct topics within these discussions and (2) *Topic Classification*: for each firm-year observation, we employ the GPT-4o API to automatically classify whether the discussions contain any of the identified topics.

We further analyze whether firms discuss the current versus planned use of AI. To achieve this, we continue to use the GPT-4o model to evaluate each AI-related discussion and classify it into intentions to adopt AI or indications that the firm has already adopted AI. AI discussions are categorized as "intention" if they indicate plans, aspirations, future possibilities, or goals to adopt, apply, or invest in AI without confirming current implementation or use. Discussions are classified as "existing action" if they confirm actual, present, or past adoption, application, or investment in AI technologies or systems.[7,8] The prompts are provided in the Online Appendix.

Determinants of AI Disclosure

As discussions of AI in 10-K reports are voluntary to some extent, our first empirical analysis models a firm's decision to discuss AI in its annual report. We use Equation (1) below to examine factors associated with the propensity of AI disclosure in 10-K filings.

$$\begin{aligned} AIDISCLOSE_{it} = {} & \beta_0 + \beta_1 SIZE_{it} + \beta_2 ORGCHG_{it} + \beta_3 COMPLEXITY_{it} + \beta_4 LOSS_{it} \\ & + \beta_5 BETA_{it} + \beta_6 UNCERTAINTY_{it} + \beta_7 NEWEQUITY_{it} + \beta_8 BIG4_{it} + \beta_9 ANALYSTS_{it} \\ & + \beta_{10} INSTOWN_{it} + \beta_{11} AIEXPOSURE_{it} + \beta_{12} AIINCIDENT_{it} + \text{Year Fixed Effects} \\ & + \text{Industry (or Firm) Fixed Effects} + \varepsilon_{it} \end{aligned} \quad (1)$$

In Equation (1), $AIDISCLOSE_{it}$ equals 1 if firm i discusses AI in 10-K report of year t and 0 otherwise. Although our baseline model examines the incidence of AI disclosure, we consider more continuous measures (e.g., the relative frequency of AI discussions) in additional tests.

In terms of determinants of *AIDISCLOSE*, we begin by considering variables that represent firms' fundamental performance. Larger firms typically have more data that AI can learn from and may have more resources to invest in AI technologies (Babina et al. 2024). As such, larger firms may be more likely to talk about AI (*SIZE*, the natural logarithm of total assets). On the other hand, larger firms may face proprietary costs relating to their use of AI to maintain competitive advantage and therefore be less likely to discuss it. *ORGCHG* represents firms' organizational changes and is a composite factor variable that includes asset growth, sales growth, leverage, and merger and acquisition activity (Dorantes, Li, Peters, and Richardson 2013). *COMPLEXITY* is another factor variable, comprising the number of

[7] Prompts used to guide GPT-4o in scoring discussions on specificity, topics, and current versus planned use are available in the Online Appendix. We keep the parameter "temperature" in the OpenAI API as 0 to ensure replicable results.

[8] To assess the reliability of the large language model (LLM)-generated outputs, we conducted a manual validation exercise. Specifically, we randomly selected a subset of observations and compared the LLM classifications with our own evaluation. The results indicate a high degree of agreement, supporting the reliability of the LLM outputs. Although LLM-based approaches enable scalable and consistent analysis of large textual datasets, they are not without limitations. LLM outputs may reflect model-specific biases arising from the training data and underlying architecture. We acknowledge this inherent limitation and interpret our results in light of this consideration. At the same time, to mitigate concerns regarding systematic bias introduced through implementation, we apply the LLM using standardized prompts and procedures across all observations. This uniform application reduces variation attributable to prompt design or execution and helps ensure that any residual bias is not driven by inconsistent treatment across the sample.

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segments, the presence of foreign transactions, and the existence of restructuring activities (Dorantes et al. 2013). *UNCERTAINTY* captures operational uncertainty in business operations and is calculated as the standard deviation of net income over the previous five years, scaled by sales (Kobelsky, Richardson, Smith, and Zmud 2008; Pincus, Tian, Wellmeyer, and Xu 2017). Firms undergoing significant changes (e.g., business expansion, taking on additional debt, and mergers and acquisitions) that are innately complex and experience operational uncertainty may find it difficult to adopt or develop AI technology and are less likely to discuss it. However, these firms may be more inclined to invest in AI because they have a greater need for it, thereby leading to more AI discussion.

LOSS is an indicator variable set to 1 if net income is negative and 0 otherwise. Firms experiencing losses may face limited resources to invest in AI. However, they may also be motivated to reassure investors by discussing AI initiatives as a way to signal that they are adopting technology to mitigate operational or market risks. *BETA* measures the sensitivity of a firm's asset returns to market returns, representing the equity beta over the last fiscal period. Firms with high *BETA* are more exposed to market risk and thus might discuss AI initiatives to signal their adaptability to volatile market conditions. On the other hand, high *BETA* firms may adopt more conservative communication strategies, avoiding speculative discussions around AI. We additionally consider *NEWEQUITY*, which is an indicator that equals 1 if a firm issues new equity or new debt in the following year and 0 otherwise (Frankel, McNichols, and Wilson 1995; Dorantes et al. 2013). Firms raising financing may discuss AI to increase their attractiveness; however, they may also avoid topics that may increase uncertainty for the firm.

We include variables that reflect external monitoring. *BIG4* is an indicator variable that equals 1 if the firm is audited by one of the Big 4 auditors and 0 otherwise. We expect that *BIG4* increases the likelihood of firms discussing AI, as Big 4 auditors often enforce stricter reporting standards and encourage more comprehensive disclosure. *ANALYSTS* is the natural logarithm of the number of analysts following the firm at the end of year *t* (Ajinkya, Bhojraj, and Sengupta 2005; Dorantes et al. 2013). We expect *ANALYSTS* to be positively related to AI discussion; AI discourse could serve as a way for firms to highlight strategic initiatives, helping analysts better understand the firm's growth plans. *INSTOWN* is the percentage of total institutional shares out of total shares outstanding. On the one hand, firms with higher institutional ownership may face more scrutiny, incentivizing them to disclose AI initiatives to meet the transparency demands of sophisticated investors. On the other hand, firms may downplay AI initiatives if they fear that these investments will be viewed as risky or uncertain by institutions.

We introduce two variables specifically related to AI. The first captures firms' exposure to AI, which includes their alignment with AI-related technologies, the relevance of AI applications to their operations, and the potential for AI to impact their competitive landscape or business model. Inspired by Chen and Srinivasan (2024), who use the similarity between the NAICS description of each industry and titles/abstracts of AI-related patents to create an industry-level AI exposure measure, we create a firm-year level AI exposure measure (*AIEXPOSURE*) by calculating the similarity between firms' Item 1 Business Description and the abstracts of AI-related patents filed up to that period. [9] The higher the *AIEXPOSURE*, the more exposed the firm is to AI technologies. We expect *AIEXPOSURE* to be positively associated with the propensity to discuss AI in 10-K reports, as these firms are more likely to use AI or have their business models impacted by AI.

Although *AIEXPOSURE* is an *ex ante* measure, our second measure captures AI exposure in an *ex post* sense: the number of AI-related incidents (*AIINCIDENT*) reported for each firm that year.[10] These incidents are mostly about AI misuse. For example, three female makeup artists working for MAC, a subsidiary of Estee Lauder, brought legal action against the company, claiming that they were wrongfully terminated after an AI tool with facial recognition exhibited bias, leading to discrimination against them based on their gender (Bargh 2022). Incidents like these are captured in the AI Incident Database maintained by the Responsible AI Collaborative. Negative publicity surrounding AI incidents may create an incentive for firms to address AI more transparently and formally in their annual reports, either to manage reputational risk or to demonstrate a proactive approach to AI governance. Therefore, we expect that a higher *AIINCIDENT* count will increase the likelihood of AI discussion in the 10-K, particularly in the Item 1A Risk Factors section.

Equation (1) is estimated using an ordinary least squares regression (OLS) with industry or firm fixed effects in alternate specifications. Firm fixed effects control for time-invariant firm characteristics, so identification in those

[9] We obtained patents from Google Patents at https://console.cloud.google.com/marketplace/details/google_patents_public_datasets/google-patents-public-data. We used the AI keywords described in Section III to identify AI-related patents, comparing each firm's Item 1 Business Description with the aggregated abstracts of all AI-related patents available up to that fiscal year, following Chen and Srinivasan (2024).

[10] AI incidents were obtained from https://incidentdatabase.ai/. The AI Incident Database (AIID) was introduced and systematically documented by McGregor (2021), who defines an AI incident as a situation in which an AI system caused, or nearly caused, real-world harm. Since its creation, the AIID has been widely adopted and cited by both academic research and authoritative industry and policy reports, supporting its external validity and relevance. Notably, the Stanford University AI Index Report, one of the most influential annual assessments of AI development and societal impact, relies on AIID data to document trends in AI-related harms and risks (Stanford University Human-Centered Artificial Intelligence (Stanford HAI) 2023, 2024, 2025).

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models relies on within-firm variation over time. Adopting a linear probability model estimated with OLS allows us to retain the full sample, in contrast to a logistic or probit regression with firm fixed effects.

Capital Market Perceptions of AI Disclosure

In our third research question, we study the capital market perceptions of AI disclosure in 10-K filings. In particular, we consider AI disclosure's impact on the quality of the firm's information environment.

Prior research has widely documented that disclosure in 10-K filings is informative to capital market participants. For example, MD&A information is associated with reduced error and dispersion in analysts' earnings forecasts (Barron, Kile, and O'Keefe 1999), whereas risk disclosures are associated with more dispersion in analysts' forecasts and increased stock return volatility (Kravet and Muslu 2013). Lehavy, Li, and Merkley (2011) find that the readability of the 10-K filing based on the Gunning Fog Index affects the dispersion and error of analyst earnings forecasts. Similarly, Loughran and McDonald (2014) document that a simpler measure of readability, the file size of the 10-K, is associated with post-filing volatility, analyst forecast error and dispersion.

Specific disclosures about AI risks and opportunities may inform capital market participants who are seeking to understand how AI will affect each firm's business model and the attendant social and legal risks. Consistent with investor interest in these disclosures, the 2024 proxy season saw a range of shareholder proposals demanding more transparency on AI issues: to name a few, the American Federation of Labor and Congress of Industrial Organizations (AFL-CIO) asked Apple to disclose its ethical guidelines around AI usage and Arjuna Capital asked Microsoft to disclose risks to its own operations and finance from AI, as well as risks to public welfare from AI-generated misinformation and disinformation (Katz and McIntosh 2024). Jia et al. (2024) document a significant increase in financial analysts' interest in generative AI, evidenced by increasing AI-related questions in conference calls.

Although the bulk of the evidence in accounting and finance literature documents that more disclosure is associated with an improved information environment and lower uncertainty for market participants (e.g., Lang and Lundholm 1996; Morgan 2002; Hope 2003; Akins 2018), how AI disclosure will affect the quality of information environment is not clear *ex ante*. On the one hand, AI disclosure may reduce uncertainty by clarifying firms' strategies for maintaining competitive advantage. On the other hand, such disclosures may increase uncertainty if they highlight risks that were previously unrecognized by investors or other financial statement users. AI disclosures might reveal the firm's exposure to competitive pressures, technological obsolescence, or the possibility that rivals could achieve first-mover advantages—concerns that create greater ambiguity about future performance. Finally, AI disclosures may have little or no effect if they are vague or boilerplate and fail to provide substantive, firm-specific information.

We model the association between AI disclosure and information environment using ordinary least squares (OLS) regression as shown Equation (2):

$$\begin{aligned} &QUALITY_INFORMATION_ENVIRONMENT_{it} \\ &\quad = b_0 + b_1 AIDISCLOSE_{it} + b_2 POSTCHATGPT_{it} + b_3 AIDISCLOSE_{it} * POSTCHATGPT_{it} + b\text{Controls}_{it} \\ &\qquad + \text{Firm Fixed Effects} + \text{Year Fixed Effects} + \varepsilon_{it} \end{aligned} \tag{2}$$

The subscripts i and t stand for firm and year. In Equation (2), $AIDISCLOSE_{it}$ equals 1 if firm i discusses AI in 10-K report of year t and 0 otherwise. OpenAI's release of ChatGPT is a breakpoint that sharply increases capital market participants' awareness of AI technology (as demonstrated by Jia et al.'s (2024) evidence on increased questioning); we hence examine capital market perceptions separately before and after this breakpoint, with $POSTCHATGPT$ set to 1 if the 10-K report was issued after the launch of ChatGPT on November 30, 2022 and 0 otherwise. The coefficient b_1 captures the difference in information environment for firms that discuss AI in their 10-K reports versus those that do not before the launch of ChatGPT (i.e., $POSTCHATGPT = 0$). The coefficient b_2 represents the difference in information environment for 10-K reports issued after the launch of ChatGPT compared with those issued before, for firms that do not discuss AI in their 10-K (i.e., $AIDISCLOSE = 0$). Lastly, the coefficient b_3, $AIDISCLOSE * POSTCHATGPT$, captures the average differential effect of AI disclosure on information environment after the launch of ChatGPT.

Following a stream of literature (most recently, Christensen, Fronk, Lee, and Nelson 2024), we capture the quality of the information environment ($QUALITY_INFORMATION_ENVIRONMENT$) with the following measures: post-filing return volatility ($POSTFILINGVOL$), analyst forecast dispersion ($DISPERSION$), and analyst forecast error ($UNEXPEARN$). As our study focuses on the informativeness of specific disclosures in 10-K filings, we adopt control variables from Loughran and McDonald (2014); Bonsall, Leone, Miller, and Rennekamp (2017); and Christensen et al. (2024). Specifically, we control for pre-filing performance ($ALPHA$), pre-filing volatility ($PREFILINGVOL$), abnormal

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returns (*ABSCAR*), book to market ratio (*BTM*), and market value of equity (*LNMVE*). We also control for firms' exposure to AI (*AIEXPOSURE*) to disentangle the effect of AI *disclosure* from AI *exposure*. For the regressions with the financial analyst outcomes (*DISPERSION* and *UNEXPEARN*), we include the number of financial analysts following a firm (*ANALYSTS*) as an additional control variable. We hypothesize a negative (positive) coefficient on b_3 if $AIDISCLOSE * POSTCHATGPT$ is associated with lower (higher) uncertainty. Because we include firm and year fixed effects, b_3 captures the difference-in-differences impact of AI disclosure on the information environment for the post versus pre-ChatGPT period.

IV. DATA AND SAMPLE

To explore the evolution of AI disclosure, we extract 49,253 10-K reports issued between January 1, 2017 and May 31, 2024 from SEC EDGAR. We choose 2017 as the starting point for our analysis because it marks a significant rise in AI adoption (McKinsey 2022; Anantharaman et al. 2024). The sample period ends on May 31, 2024, as we collected the data in June 2024. We exclude 12,302 observations that are not covered by Compustat, along with an additional 1,559 records with missing values for key financial variables, including total assets, total revenue, and net income. This leaves us with 35,392 firm-year observations from 6,937 unique firms for the descriptive analysis.

We obtain most variables used for the determinants and perceptions models from Compustat, CRSP, Audit Analytics, and I/B/E/S. Variable definitions and data sources are provided in Appendix A. After removing observations missing values necessary for the determinants model, we have 15,837 observations to analyze determinants of AI disclosure. In the perceptions test, as we aim to study the association between AI disclosure and information environment pre- and post-ChatGPT, this model is estimated on data from 2021 to 2023, roughly one year before and after the launch of ChatGPT. After removing observations missing values necessary for the perceptions model and retaining those from 2021 to 2023, we have 6,876 observations for the analysis of perceptions of AI disclosure. Table 1 outlines the sample derivation for the analyses of evolution, determinants, and perceptions of AI disclosure.

V. EMPIRICAL RESULTS

Evolution of AI Disclosure

Distribution

The most common sections that feature AI discussion are Item 1 Business Description, Item 1A Risk Factors, and Item 7 MD&A. These three sections account for approximately 85 percent of the overall AI discussion. Thus, we focus our analysis on these three sections.[11] Figure 1 shows the overlap among the three sections that contain AI discussions. About 80 percent (1,220 out of 1,502) of the observations that discuss AI in the MD&A also mention AI in Business Description, and about half (1,348 out of 2,755) of the observations that discuss AI in Risk Factors also mention AI in Business Description.

Figure 2 provides a visual representation of the yearly and quarterly distribution of AI keyword mentions in Business Description, Risk Factors, and MD&A. A nontrivial proportion of the sample observations—around 14.6 percent—were discussing AI in Business Description even prior to ChatGPT's release (the last such disclosures were made in the calendar year 2022), with 7.4 percent (5.5 percent) making references in Risk Factors (MD&A). In the two years post-ChatGPT issuance, discussion in Business Description (MD&A) increases to 23.23 percent (9.82 percent) of the sample, with Risk Factors discussion now found in almost 34 percent of the sample. From Figure 2, Panel A, it is evident that AI discussion in Business Description and MD&A increases quite gradually over the sample period, with AI discussion in Risk Factors spiking sharply post-ChatGPT's release in late 2022. From Figure 2, Panel B, we further isolate spikes in Risk Factors mentions to key events beyond the release of ChatGPT, such as the open letter about AI's profound risks to society signed by Elon Musk and others on March 29, 2023 (Metz and Schmidt 2023) and the issuance of the White House Executive Order for safe AI on October 23, 2023 (Biden 2023).

Table 2 presents the industry distribution of AI mentions in Business Description, Risk Factors, and MD&A, categorized using the Fama-French 12 industry classification. Across all three sections, the Technology sector consistently exhibits the highest frequency of AI mentions, as expected. Interestingly, other sectors have reasonable rates of AI discussion; firms in Manufacturing and Consumer Durables (e.g., cars, televisions (TVs), furniture, and household appliances) discuss AI relatively more frequently in Business Description. Firms in Consumer Durables and "Other" (e.g., mines, construction, transportation, hotels, and entertainment), discuss AI with notable frequency in Risk Factors.

[11] All firms that mention AI keywords in any of the other sections also mention AI in these three sections.

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TABLE 1

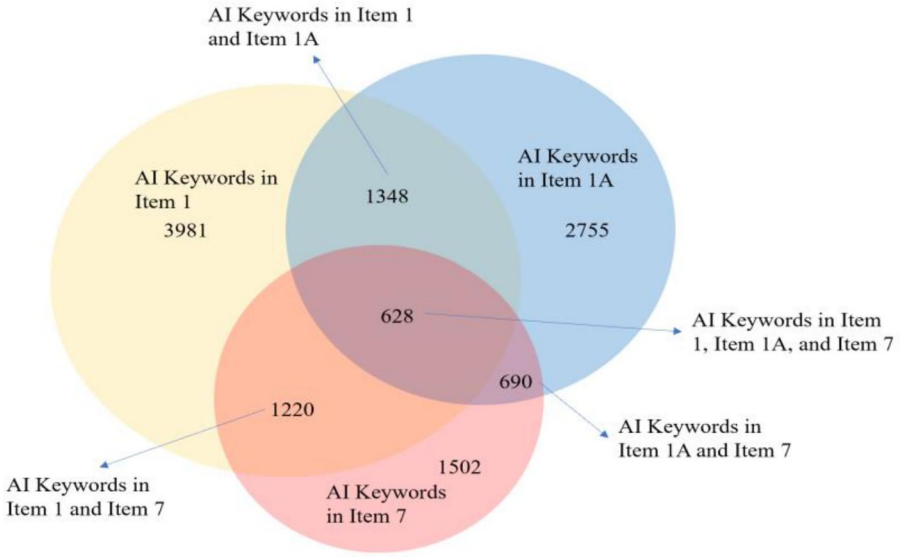
Sample Derivation

10-Ks issued from 01012017 to 05312024	49,253				
Less: observations not covered by Compustat	−12,302				
Less: observations with missing total assets, total revenue, and net income variables	−1,559				
Total sample to examine the evolution of AI disclosure (firm-years)		**35,392**			**35,392**
	Less: observations with missing values for variables in the determinants model	−19,555	Less: observations with missing values for variables in the consequence model	−20,737	
			Less: observations outside of 2021 and 2023	−7,779	
	Total sample to examine the determinants of AI disclosure (firm-years)	**15,837**	**Total sample to examine the consequence of AI disclosure (firm-years)**		**6,876**

This table shows the sample derivation for the evolution of AI disclosure, the determinants of AI disclosure, and the consequence of AI disclosure analyses.

FIGURE 1

Overlap Among Item 1, Item 1A, and Item 7 That Contains AI-Related Keywords



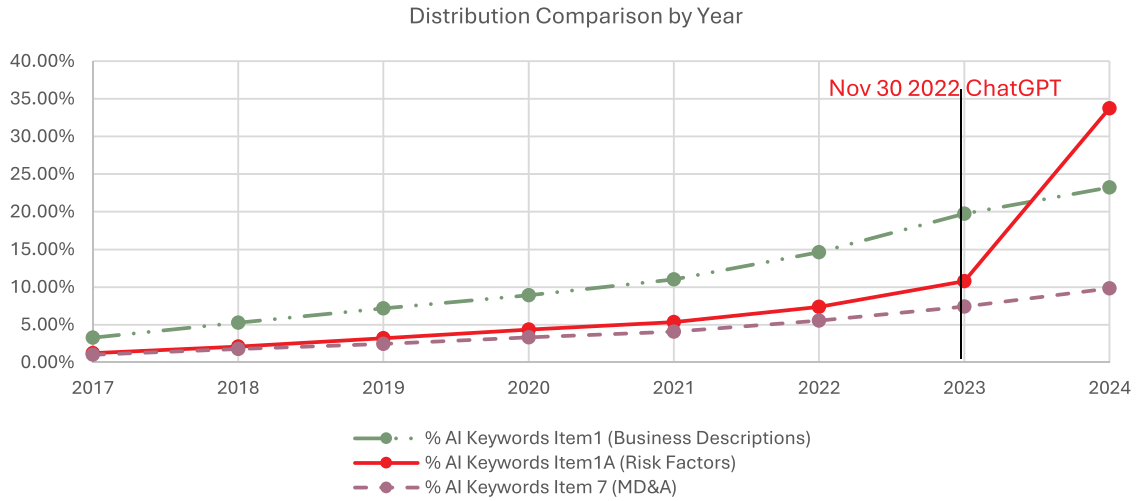
Item 1 represents Business Description, Item 1A represents Risk Factors, and Item 7 represents MD&A.
(The full-color version is available online.)

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FIGURE 2

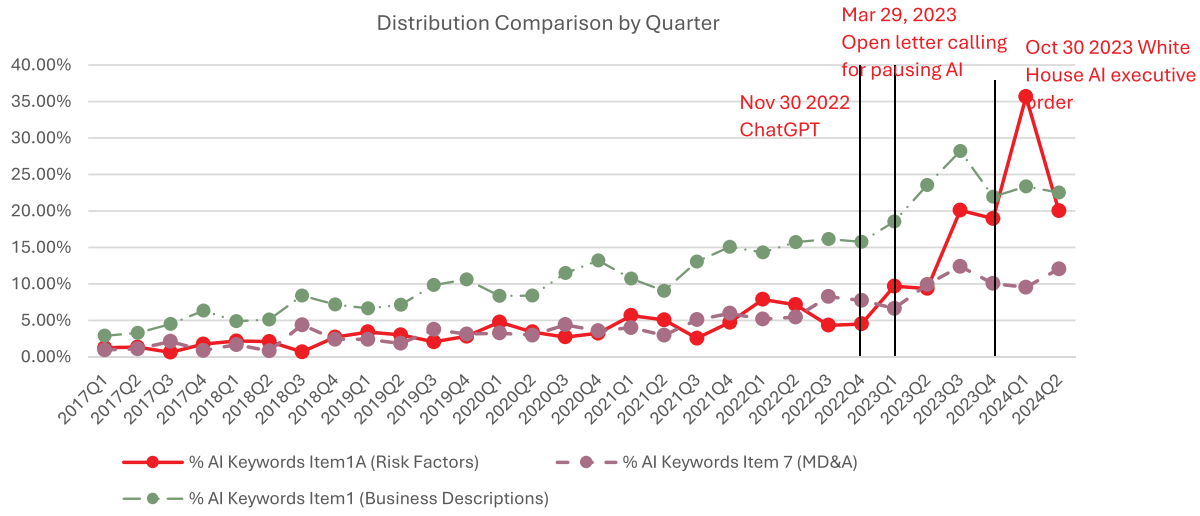
Distribution of AI Keyword Mentions in 10-K Reports Over Time

Panel A: Yearly Distribution of AI Keyword Mentions



This figure shows the percentage of 10-K reports that contain AI keywords in individual sections over the years.

Panel B: Quarterly Distribution of AI Keywords Mentions



This figure shows the percentage of 10-K reports that contain AI keywords in individual sections over the quarters. (The full-color version is available online.)

Lastly, in the MD&A section, firms in Manufacturing and Healthcare sectors exhibit relatively higher frequency of AI mentions. Utilities and Chemicals show the lowest incidence of AI discussion in all these sections.

We next analyze the frequency of AI keyword appearances in Business Description, Risk Factors, and MD&A, with Figure 3, Panel A visualizing the trend of average frequency of AI keywords by year. We observe that AI keywords consistently appear more frequently in Risk Factors than in MD&A, and in 2024, the frequency in Risk Factors exceeds that in Business Description. In all sections, the terms "AI" or "artificial intelligence" are the most frequently mentioned, followed by "machine learning" and "generative AI," as depicted in Figure 3, Panel B.

Figure 4, Panel A visually represents the number of firms that mention AI in Business Description, Risk Factors, and MD&A for the first time each year. Notably, (1) more than 100 firms make first-time mentions of AI in their

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TABLE 2

Industry Distribution of AI Disclosure

Industry Label	AI Disclosure in Item 1 (Business Description)			AI Disclosure in Item 1A (Risk Factors)		AI Disclosure in Item 7 (MD&A)	
	#	#	%	#	%	#	%
Finance	7,042	313	4.44	448	6.36	112	1.59
Healthcare	7,039	628	8.92	407	5.78	242	**3.44**
Technology	5,443	1,826	**33.55**	946	**17.38**	749	**13.76**
Others	4,351	430	9.88	366	**8.41**	148	3.40
Wholesale and Retail	2,656	212	7.98	156	5.87	43	1.62
Manufacturing	2,601	284	**10.92**	120	4.61	97	**3.73**
Energy	1,395	50	3.58	48	3.44	27	1.94
Utilities	1,311	15	1.14	39	2.97	16	1.22
Consumer Nondurables	1,300	66	5.08	92	7.08	21	1.62
Chemicals	866	30	3.46	23	2.66	12	1.39
Consumer Durables	793	89	**11.22**	66	**8.32**	25	3.15
Telecommunication	595	38	6.39	44	7.39	10	1.68

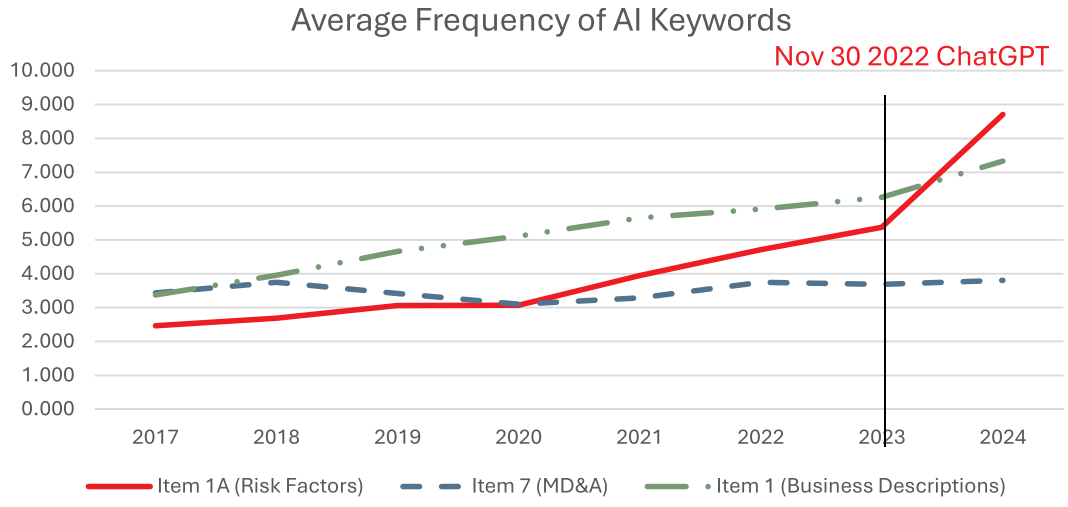
We use the Fama-French 12 industries categorization obtained from https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/Data_Library/det_12_ind_port.html. Technology includes computers, software, and electronic equipment. Consumer Nondurables include food, tobacco, textiles, apparel, leather, and toys. Consumer Durables include cars, TVs, furniture, and household appliances. Others include mines, construction, transportation, hotels, and entertainment.

Business Description every year pre-ChatGPT introduction (i.e., up to 2022); (2) first-time mentions increase over time, but not dramatically so, post-ChatGPT introduction (i.e., in 2023 and 2024 compared with 2022); (3) fewer firms mention AI in their Risk Factors pre-ChatGPT (a minimum of 41 to a maximum of 134 new firms each year; (4) that number jumps dramatically in 2024 to 872 new firms; and (5) MD&A mentions start out similar to Risk Factor mentions,

FIGURE 3

AI Keyword Frequency in 10-K Reports

Panel A: Average of AI Keyword Frequency in AI Disclosures Over Time



This figure shows the frequency of AI keywords in individual sections over the years.

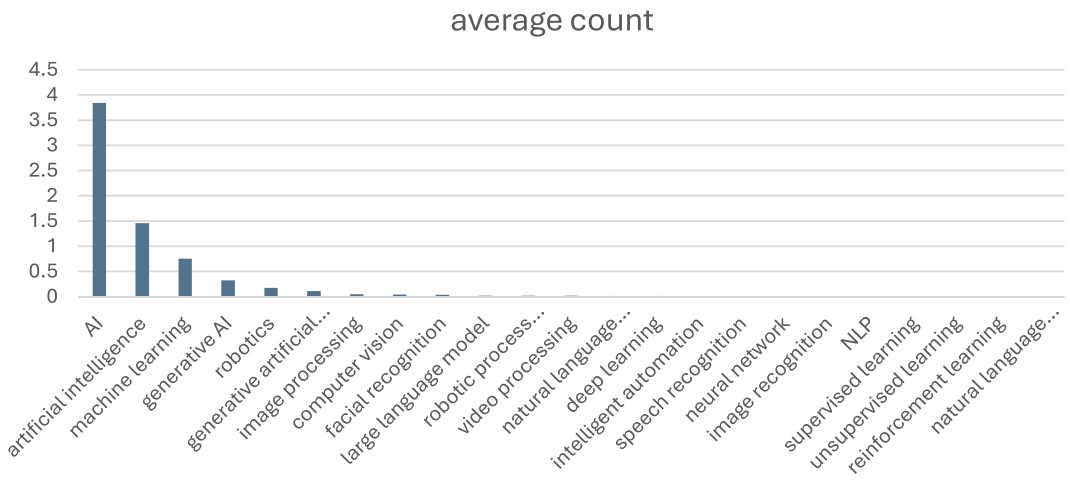
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FIGURE 3 (continued)

Panel B: Average Frequency of Each AI Keyword



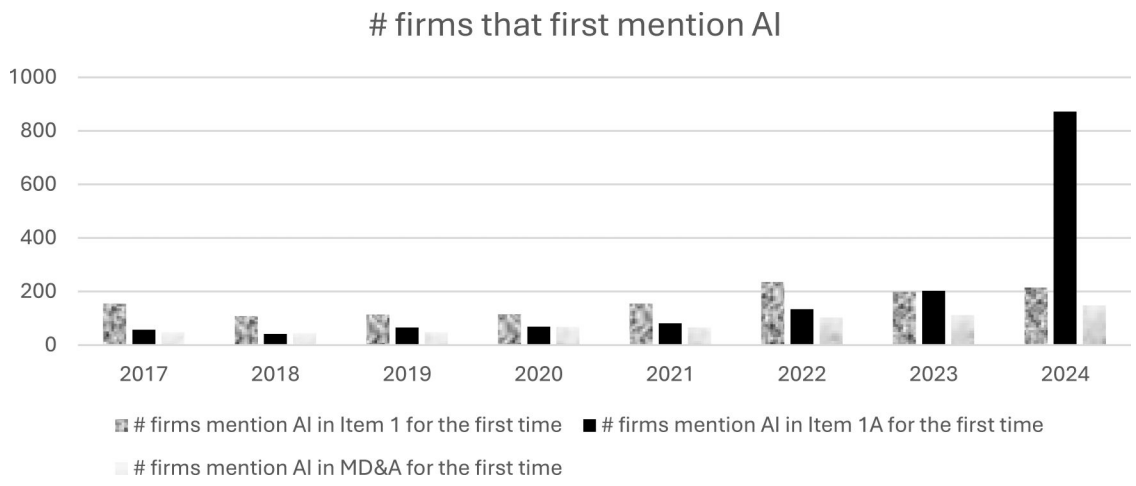
This figure shows the average frequency of each AI keyword.
(The full-color version is available online.)

with a smaller increase in recent years. Figure 4, Panel B provides a breakdown of first-time AI disclosures by industry and year. Although all sectors experienced an increase in first-time AI disclosures from 2023 to 2024, the surge was particularly pronounced in the Healthcare and Finance sectors. These two sectors, with many administrative tasks and data-based decision-making ripe for automation and optimization, are frequently mentioned as prime candidates for AI-based transformation (Qualtrics 2026).

FIGURE 4

Firms Mentioning AI for the First Time

Panel A: Number of Firms That Discuss AI for the First Time by Year



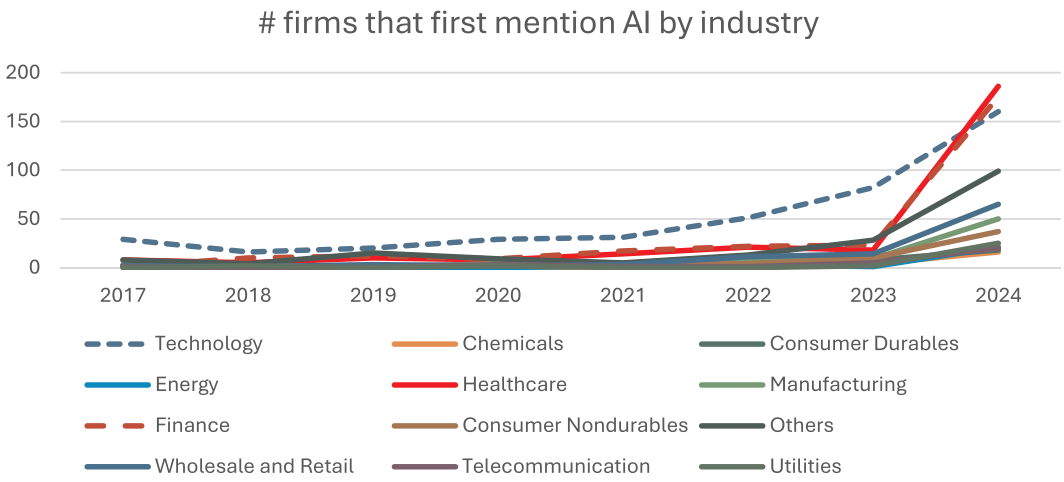
This figure shows the number of firms that discuss AI for the first time over the years.

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FIGURE 4 (continued)

Panel B: Number of Firms That Discuss AI for the First Time by Industry and Year



This figure shows the number of firms that discuss AI for the first time by industry-year, using the Fama-French industry classification. (The full-color version is available online.)

Overall, the distributional results indicate that firms are increasingly discussing AI in Business Description, Risk Factors, and MD&A, with the increase in AI mentions particularly evident in Risk Factors.

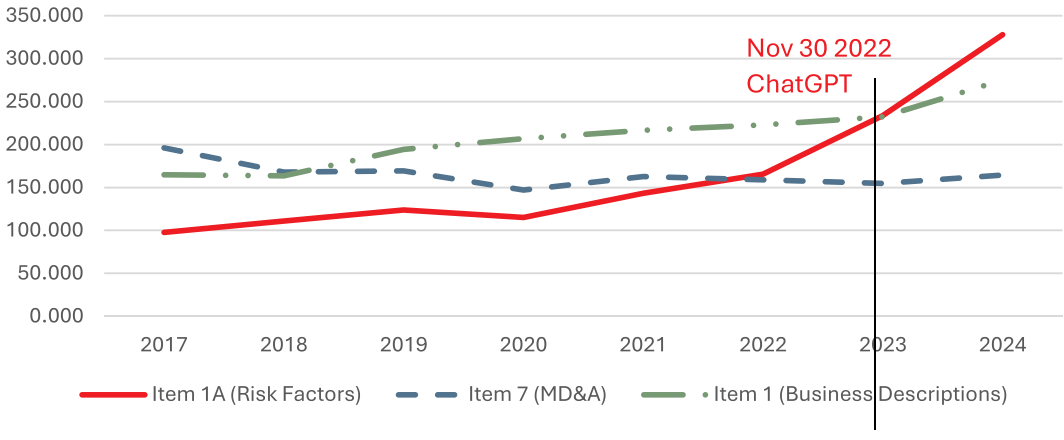
Length of AI Discussions

Figure 5, Panel A (Panel B) illustrates the average word count (average relative length, which is equal to the proportion of the AI discussion word count relative to the total word count of the section in which they appear) of AI discussions in the three sections over time. We observe a clear trend of increasing absolute and relative length of AI discussion within Business Description and Risk Factors since 2022, whereas the length of AI discussions in MD&A shows no distinct pattern.

FIGURE 5

Length of AI Discussions

Panel A: Average Word Count of AI Discussions Over Time



This figure shows the average word count of AI sentences over the years.

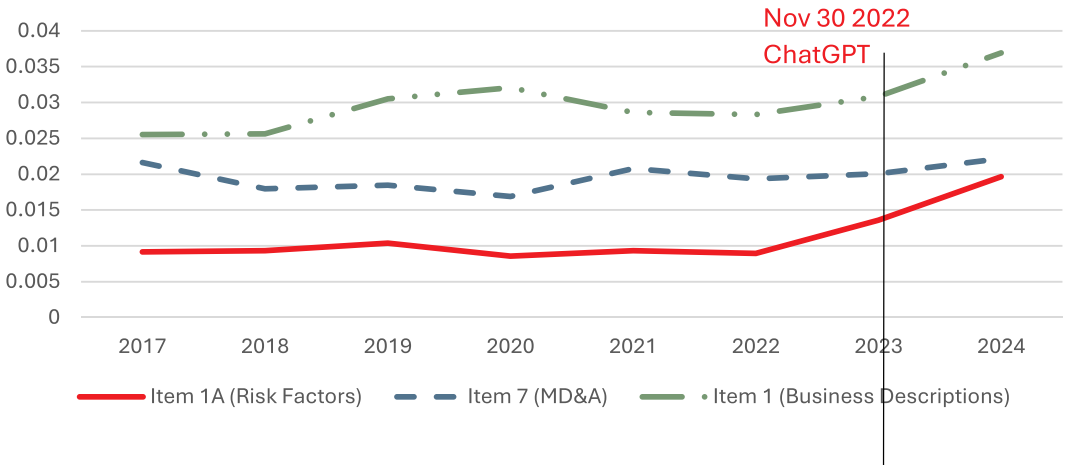
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FIGURE 5 (continued)

Panel B: Average Relative Length of AI Discussions Over Time



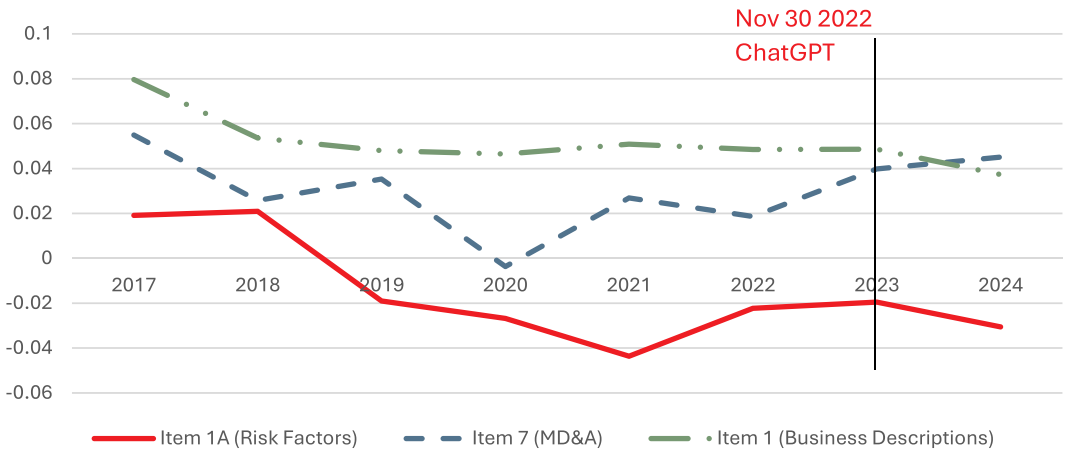
This figure shows the average proportion of AI sentences to all sentences in the corresponding section of the 10-K.
(The full-color version is available online.)

Sentiment of AI Discussions

Figure 6 illustrates the average sentiment of AI discussions over the years. The sentiment score for all AI discussions falls between −0.06 and 0.08, which is neither significantly positive nor negative; however, some noteworthy patterns exist. Over time, the average sentiment of AI discussions in Business Descriptions is quite stable. In contrast, AI discussions in Risk Factors are increasingly negative in tone, whereas those in MD&A are increasingly positive.

FIGURE 6

Average Sentiment of AI Discussions Over Time



This figure shows the average sentiment of AI sentences over the years.
(The full-color version is available online.)

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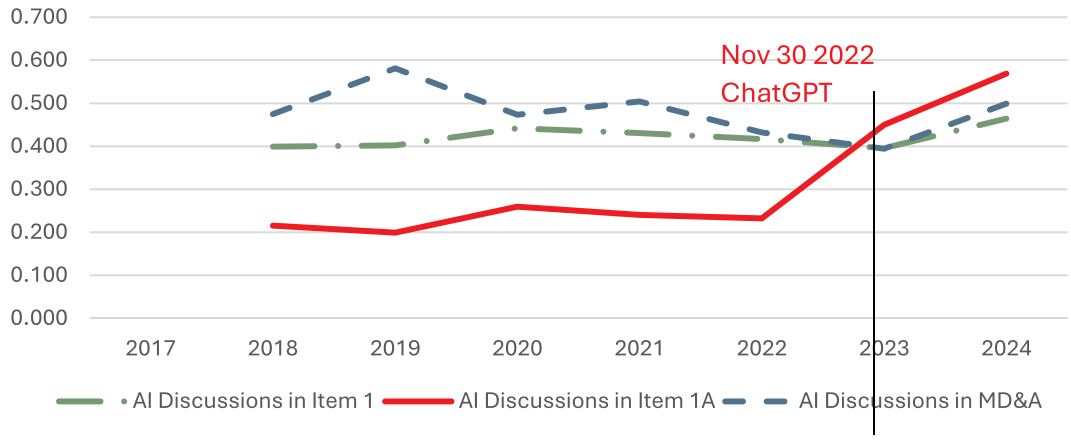
Informational Changes in AI Discussions

Figure 7, Panel A (Panel B), visualizes the average change in information of AI disclosures from a time-series (cross-sectional) perspective. From Figure 7, Panel A, AI discussions in Risk Factors have contained significantly more changes in information since 2023, as reflected in the time-series measure capturing year-on-year dissimilarity. In contrast, AI discussions in Business Description and MD&A show a decline in informational change in 2023 and an increase in 2024. AI-related discussions in Risk Factors could be becoming more firm-specific, consistent with the SEC's increasing focus on eliciting meaningful and "tailored" disclosures from issuers. From Figure 7, Panel B, we observe that, unlike the time-series measure, the cross-sectional measure, which compares a firm's AI disclosure with those of its industry-year peers, does not show significant shifts over time. This stability likely reflects the cross-sectional dissimilarity measure's ability to smooth out variations due to temporal factors affecting all firms.

FIGURE 7

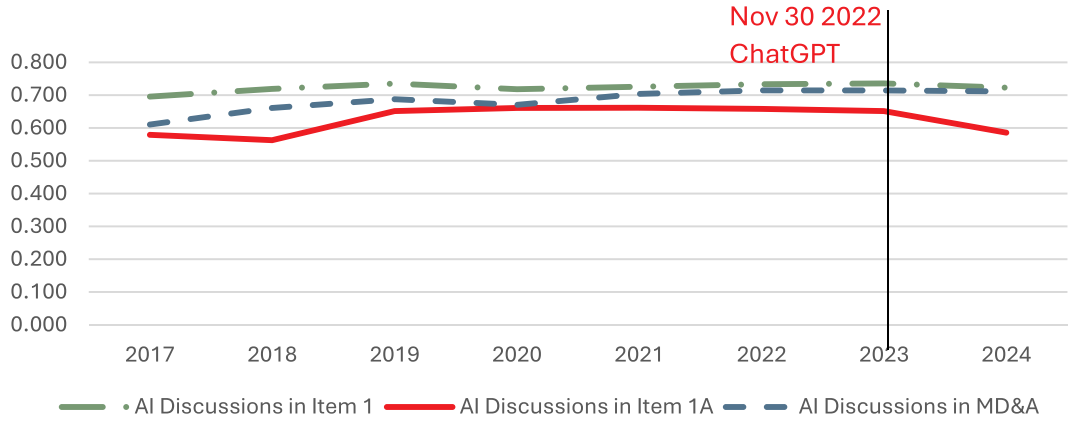
Informational Change of AI Discussions

Panel A: Average of Time-Series Informational Change of AI Discussions



This figure shows the average informational change in AI sentences over the years. The *y*-axis represents the Euclidean distance between the document embeddings from the previous year and the current year.

Panel B: Average of Cross-Sectional Informational Change of AI Discussions



This figure shows the average informational change in AI sentences across industries and over the years. The *y*-axis is the Euclidean distance between the embedding of a firm's AI-related discussion in year *t* and the average embedding of AI-related discussions from other firms within the same two-digit SIC in the same year.
(The full-color version is available online.)

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Specificity of AI Discussions—Evidence from Large Language Model Analysis

We use GPT-4o to capture the specificity of AI-related discussion. Figure 8, Panel A visualizes the trend of average specificity over time. Notably, although AI discussions in Risk Factors were the least specific prior to 2024, their specificity improved significantly in 2024, surpassing that of the MD&A section. Figure 8, Panels B–D provide a detailed breakdown of specificity categories by section. Consistent with our observations, Figure 8, Panel C highlights a sharp decrease in the "General" category for AI discussions in Risk Factors over the years, accompanied by a notable increase in the "Somewhat Specific" and "Specific" categories starting in 2022. In contrast, the distribution of specificity categories for the Business Description and MD&A sections has remained relatively stable over time.

What Are Firms Saying—Evidence from Large Language Model Analysis

AI-related topics in business description. For AI-related discussions within Business Description, we use GPT-4o to identify major topics of disclosure. Five major topics were identified: enhance products and services, improve operational efficiency and automation, support marketing and customer engagement, enhance risk management and decision support, and accelerate research and development (R&D).

Figure 9, Panel A shows the percentage of AI-related discussions in Business Description that address specific topics. The most frequent AI-related opportunity topic is enhancing product or services, i.e., how AI is used to improve the quality, personalization, or features of products or services. The next most frequently discussed is improving operational efficiency and automation, i.e., using AI to automate operational processes, optimize workflows, or provide predictive maintenance. Firms also discuss AI's role in accelerating R&D, where AI is used in research or innovation efforts, including drug discovery, data-driven prototyping, and AI-augmented design tools.

AI-related topics in risk factors. We identify ten AI-related risk topics in the Item 1A Risk Factors: technological risks, financial risks, operational risks, regulatory and legal risks, security and privacy risks, ethical and social risks, reputational damage risks, talent acquisition risks, market competition and acceptance risks, and business strategy risks.

Figure 9, Panel B shows the percentage of firms that discuss specific topics. The most frequently discussed risk is security and privacy risks, which include vulnerabilities to AI-targeted cyberattacks, the misuse of AI by bad actors (e.g., cyberattacks, data theft), and challenges in detecting and preventing sophisticated AI-driven threats. The next most common risk is reputational damage, referring to brand harm from ethical issues or controversies surrounding AI, along with negative public perception that could affect business reputation and customer trust. Following this, regulatory and legal risks are frequently discussed, focusing on compliance with evolving AI-related laws, adherence to multiple regulatory frameworks (international, federal, state, and local), heightened regulatory scrutiny, potential legal liabilities, and intellectual property concerns related to AI technologies. Companies also address ethical and social risks. Financial risks are often mentioned, referring to the high costs of AI development with uncertain returns and potential losses from failed AI integrations. Business strategy risks—i.e., from AI technologies disrupting established business models—are only discussed by less than 10 percent of all firms.[12] Furthermore, as shown in Figure 9, Panel C, discussions of AI's role in improving business operations are most prevalent, followed by enhancing products or services and growth of business discussion.

Current Versus Planned Use of AI—Evidence from Large Language Model Analysis

We use GPT-4o to identify whether AI-related discussions are implying the intention to adopt AI in the future or are providing evidence of firms having already adopted AI. Table 3 summarizes the proportion of firms discussing current versus planned use of AI in the AI-related discussions of each of the three sections. In Business Description, 72.9 percent of AI discussions reference existing AI adoption, whereas 15.2 percent discuss intentions to adopt AI. A similar pattern is observed in MD&A, with 61.7 percent indicating existing adoption and 19.4 percent indicating intention. In contrast, AI discussions in the Risk Factors section show a much lower proportion indicating existing adoption (34.3 percent) and a higher proportion suggesting intentions to adopt AI (27.2 percent).

Determinants of AI Disclosure

Table 4, Panel A presents the summary statistics of the variables used in the analysis of determinants of AI disclosure. Table 4, Panel B presents the multivariate regression results under two alternative fixed effects structures: year

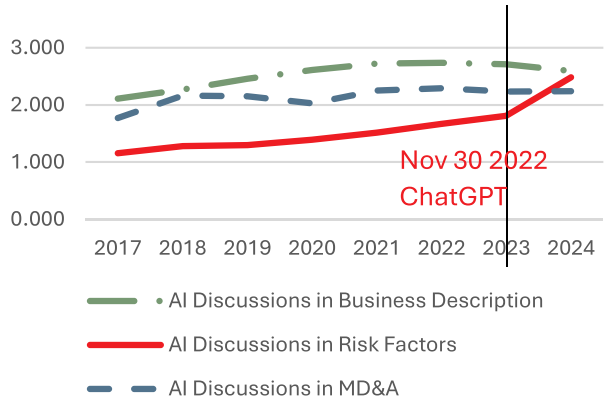
[12] Interestingly, the most discussed and least discussed of the risk topics have both taken center stage in recent public conversation. On security and privacy risks, frontier AI models released in early 2026 have been shown to have the capability to autonomously identify (and exploit) vulnerabilities in major operating systems, browsers, and legacy software, identifying some bugs that have escaped human review and detection for decades (e.g., the Mythos model released by Anthropic). On business strategy risks, the software sector faces heavy scrutiny over concerns that legacy software will be rendered obsolete by AI-native applications. See, e.g., Markman (2026).

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FIGURE 8

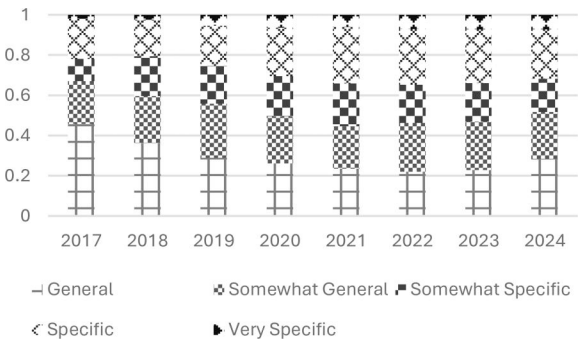
Specificity of AI Discussions

Panel A: Average Specificity of AI Discussions



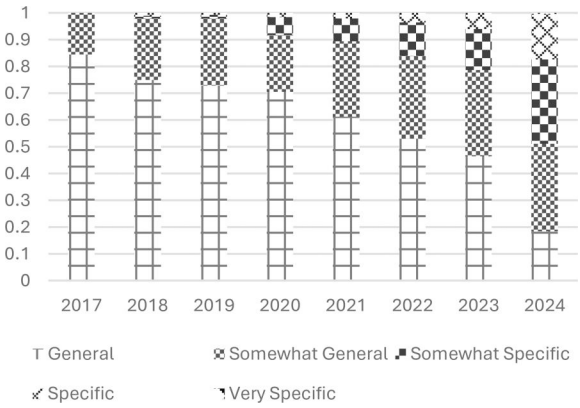
This figure shows the average specificity of AI sentences.

Panel B: Categories of Specificity for AI Discussions in Business Descriptions



This figure shows the proportion of categories of specificity for AI sentences in Item 1—Business Description.

Panel C: Categories of Specificity for AI Discussions in Risk Factors



This figure shows the proportion of categories of specificity of AI sentences in Item 1A—Risk Factors.

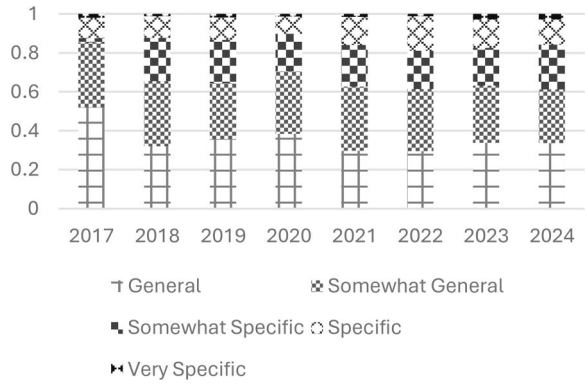
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FIGURE 8 (continued)

Panel D: Categories of Specificity for AI Discussions in MD&A



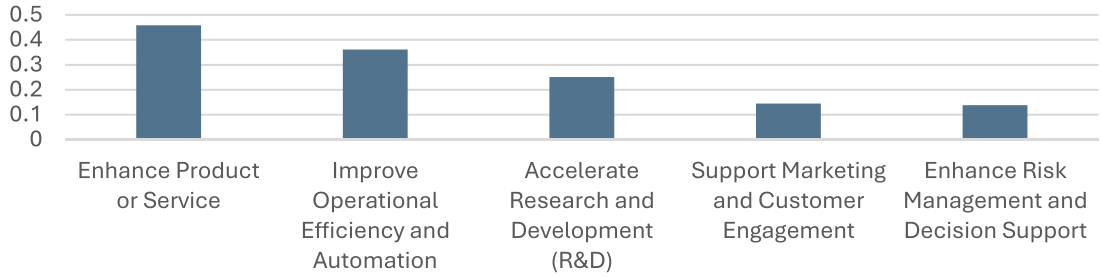
This figure shows the proportion of categories of specificity of AI sentences in Item 7—MD&A.
(The full-color version is available online.)

and industry fixed effects and year and firm fixed effects. These alternative structures allow us to explore whether the decision to disclose AI is shaped more by cross-sectional factors or firm-specific factors. Incorporating firm fixed effects significantly improves the explanatory power of the model, as measured by the adjusted R^2, indicating the presence of firm-specific, time-invariant factors associated with the AI disclosure decision.

FIGURE 9

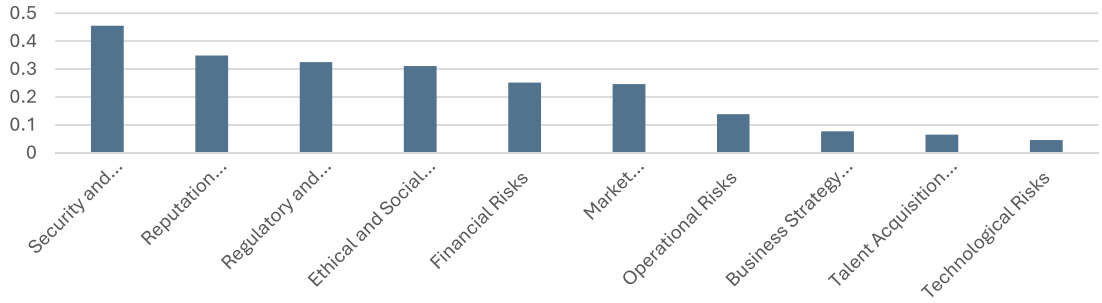
Main Topics of AI Discussions

Panel A: Distribution of Item 1 AI Discussion Main Topics



This figure shows the percentage of AI sentences that cover each individual topic in Item 1.

Panel B: Distribution of Item 1A AI Discussion Main Topics



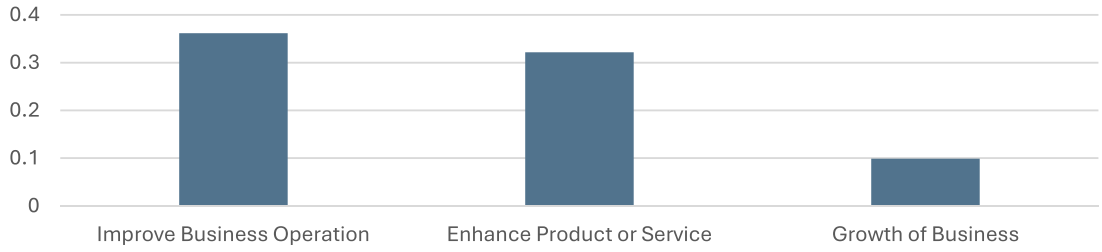
This figure shows the percentage of AI sentences that cover each individual topic in Item 1A.

(continued on next page)

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FIGURE 9 (continued)

Panel C: Distribution of Item 7 AI Discussion Main Topics



This figure shows the percentage of AI sentences that cover each individual topic in Item 7A.
(The full-color version is available online.)

TABLE 3

Summary Statistics Intention or Existing Action of AI Adoption

	Item 1 AI Discussions (Business Description)		Item 1A AI Discussions (Risk Factors)		Item 7 AI Discussions (MD&A)	
Disclose Year	**Intention**	**Existing Action**	**Intention**	**Existing Action**	**Intention**	**Existing Action**
Count	3,981	3,981	2,755	2,755	1,502	1,502
Mean	0.152	0.729	0.272	0.343	0.194	0.617
Std. dev.	0.360	0.444	0.445	0.475	0.396	0.486
Min	0	0	0	0	0	0
25%	0	0	0	0	0	0
50%	0	1	0	0	0	1
75%	0	1	1	1	0	1
Max	1	1	1	1	1	1

This table shows the proportion of firms discussing intention or existing action of AI adoption in Item 1, Item 1A, and Item 7.

Larger firms (*SIZE*) are more likely to disclose AI, particularly in the Business Description and Risk Factor sections. Firms experiencing more growth (*ORGCHG*) and more complex firms (*COMPLEXITY*) are more likely to disclose AI, particularly in Business Description and MD&A sections, consistent with greater AI investments in operations. Loss-making firms (*LOSS*) are more inclined to discuss AI in the Business Description and Risk Factors sections, whereas those with greater sensitivity to market risk (*BETA*) are more likely to disclose AI across all three sections. Interestingly, firms with volatile earnings (*UNCERTAINTY*) make fewer mentions of AI in Business Description and MD&A, and firms issuing new equity (*NEWEQUITY*) are less inclined to discuss AI in Business Description and Risk Factors sections.

Firms audited by Big 4 auditors (*BIG4*) exhibit a marginally higher propensity to discuss AI, although this is confined to Risk Factors. Firms followed by more analysts (*ANALYSTS*) are more inclined to discuss AI in Risk Factors and MD&A. In a surprising finding, firms with higher levels of institutional ownership (*INSTOWN*) are less likely to discuss AI.

Both our AI-related measures are highly relevant: *ex ante* AI exposure, as measured by the similarity of a firm's Business Description section to AI-related patents filed to date (*AIEXPOSURE*), is highly significant across sections, and firms with more AI-related incidents (*AIINCIDENTS*) are more likely to address AI in Risk Factors and MD&A, likely reflecting an effort to manage reputational and operational risks arising from these incidents.[13]

[13] In Online Appendix, Figure OA.1, we extract the year fixed effects coefficients from Equation (1), estimated with industry fixed effects (results are similar when using firm fixed effects) from models with each of the four dependent variables (*AIDISCLOSE*, *AIBUSINESS*, *AIRISK*, and *AIMDA*). The plot reveals a clear upward trend in AI disclosure across all four disclosure measures from 2018 to 2023, with a notable spike in 2022, which coincides with the emergence of ChatGPT and growing public attention to generative AI. These patterns are consistent with the descriptive results shown in Figures 2, 3, and 5.

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TABLE 4

Determinants of AI Disclosures

Panel A: Summary Statistics of the Variables Used in the Determinants Model

Variable	Obs.	Mean	Std. Dev.	Min	Max
AIDISCLOSE	15,837	0.1985	0.3989	0.0000	1.0000
AIBUSINESS	15,837	0.1674	0.3733	0.0000	1.0000
AIRISK	15,837	0.0687	0.2530	0.0000	1.0000
AIMDA	15,837	0.0629	0.2428	0.0000	1.0000
SIZE	15,837	6.4510	2.4966	−5.8091	12.8471
ORGCHG	15,837	0.0237	0.9531	−6.1551	22.0859
COMPLEXITY	15,837	0.3219	1.2834	−1.1645	4.6366
LOSS	15,837	0.4274	0.4947	0.0000	1.0000
BETA	15,837	1.0633	0.4784	−0.9429	3.1437
UNCERTAINTY	15,837	5.4121	35.5302	0.0008	772.4720
NEWEQUITY	15,837	0.5581	0.4966	0.0000	1.0000
BIG4	15,837	0.6541	0.4757	0.0000	1.0000
ANALYSTS	15,837	1.4510	1.2133	0.0000	3.9703
INSTOWN	15,837	0.4991	0.3952	0.0000	1.5359
AIEXPOSURE	15,837	0.1792	1.0742	−0.9657	3.2100
AIINCIDENT	15,837	0.0073	0.2389	0.0000	17.0000

This table shows the summary statistics for the determinants of AI disclosure analysis. Sample period for this analysis is 2017–2024. Variable definitions are provided in Appendix A.

(continued on next page)

Adding firm fixed effects adds considerable explanatory power across the board—going from 8 to 21 percent in the cross-sectional models to 56–69 percent with firm fixed effects. Many of the cross-sectional factors lose statistical significance after adding firm fixed effects, but *SIZE* and *AIEXPOSURE* remain consistently significant. In untabulated analysis, we attempt to unpack the firm-level traits embedded in the fixed effects; when we augment the cross-sectional models (only year and industry fixed effects) with measures of prior-year AI disclosure, the explanatory power jumps to levels comparable to those of the firm fixed effects models. Augmenting the firm fixed effects models with prior-year AI disclosure improves the explanatory power further but only modestly so. We conclude from these comparisons that there is strong evidence of persistence in AI disclosure; intuitively, once firms begin discussing AI, they tend to continue doing so. We then attempt to understand what drives first-time AI disclosure; on re-estimating Equation (1) for first-time AI disclosers only, *SIZE* and *AIEXPOSURE* remain significant determinants of first-time AI disclosure. In other words, as technologies develop in a direction that intersects with (or potentially disrupts) the firm's business, we are likely to see AI-related discussion.

Additional Analyses and Specification Checks

Our analysis of determinants currently examines only the incidence of AI disclosure as a dependent variable; as our descriptive discussion shows, firms vary widely in how substantive and lengthy their AI disclosures are. To distinguish between firms that could be merely signaling AI engagement (with a short or vague mention) and those that discuss AI more substantively, we expand out the determinants analysis with three more continuous measures of AI disclosure: the intensity of AI discussion (number of AI-related keywords) relative to the length of the relevant section in 10-K, the number of AI topics discussed (captured using GPT-4o), and the specificity of AI discussion (captured using GPT-4o).[14] Synthesizing across the many specifications, certain variables, such as *SIZE*, *ANALYSTS*, *INSTOWN*, and *AIEXPOSURE*, are consistently associated with AI disclosure across both industry and firm fixed

[14] A semantic similarity disclosure measure, benchmarking a firm's AI narrative against narrative about risk or strategy, is another potentially useful measure. However, such a score can only be computed for firm-years with AI disclosure. Therefore, to retain non-AI disclosers in the analysis, we would have to replace missing values with zeroes for those firms, conflating "no AI disclosure" with "zero similarity." For this reason, we do not include this measure in our main tests. That said, examining semantic similarity within the subsample of AI disclosers represents a promising avenue for future research.

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TABLE 4 (continued)

Panel B: Determinants of AI Disclosures

		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
		AIDISCLOSE		*AIBUSINESS*		*AIRISK*		*AIMDA*	
		Coeff. (t-stats)							
SIZE	+	0.0124***	0.0161**	0.0095***	0.0148**	0.0104***	0.0195***	0.0011	0.0043
		(3.65)	(2.03)	(2.97)	(2.05)	(5.04)	(3.67)	(0.48)	(0.71)
ORGCHG	?	0.0104***	−0.0033	0.0087***	−0.0059**	0.0027	−0.0019	0.0066**	0.0013
		(3.09)	(−1.27)	(2.88)	(−2.44)	(1.36)	(−1.56)	(2.55)	(0.79)
COMPLEXITY	+	0.0133***	0.0017	0.0112**	−0.0001	0.0037	−0.0046	0.0076**	−0.0016
		(2.61)	(0.36)	(2.34)	(−0.03)	(1.15)	(−1.20)	(2.47)	(−0.46)
LOSS	?	0.0322***	−0.0060	0.0303***	−0.0106*	0.0274***	0.0049	0.0097	−0.0028
		(3.18)	(−0.91)	(3.11)	(−1.70)	(4.20)	(1.14)	(1.49)	(−0.58)
BETA	?	0.0450***	0.0105*	0.0401***	0.0074	0.0312***	0.0053	0.0197***	−0.0015
		(5.14)	(1.68)	(4.78)	(1.30)	(5.24)	(1.09)	(3.30)	(−0.35)
UNCERTAINTY	−	−0.0001**	0.0001	−0.0002***	0.0001	−0.0000	0.0000	−0.0001***	0.0000
		(−1.97)	(1.11)	(−3.07)	(1.47)	(−0.54)	(0.75)	(−3.21)	(0.07)
NEWEQUITY	?	−0.0339***	0.0018	−0.0415***	−0.0027	−0.0174***	0.0030	−0.0092	−0.0014
		(−3.61)	(0.29)	(−4.62)	(−0.48)	(−2.95)	(0.71)	(−1.49)	(−0.30)
BIG4	+	0.0148	0.0031	0.0121	−0.0041	0.0176*	−0.0130	0.0038	0.0088
		(1.06)	(0.15)	(0.93)	(−0.23)	(1.87)	(−0.88)	(0.43)	(0.60)
ANALYSTS	+	0.0228***	0.0060	0.0215***	0.0049	0.0119***	0.0061*	0.0116***	0.0053*
		(4.00)	(1.17)	(3.93)	(0.97)	(3.06)	(1.73)	(3.29)	(1.76)
INSTOWN	?	−0.0668***	0.0005	−0.0691***	−0.0109	−0.0548***	−0.0304**	−0.0407***	−0.0092
		(−3.58)	(0.02)	(−3.76)	(−0.56)	(−4.34)	(−2.08)	(−3.06)	(−0.80)
AIEXPOSURE	+	0.0665***	0.0501***	0.0735***	0.0568***	0.0240***	0.0075**	0.0239***	0.0050
		(11.49)	(7.23)	(13.21)	(7.89)	(6.33)	(2.02)	(6.09)	(1.32)
AIINCIDENT	+	0.0177	0.0281	0.0197	0.0187	0.0350**	0.0166	0.0376**	0.0241
		(0.91)	(1.15)	(1.02)	(0.98)	(2.25)	(0.94)	(2.47)	(1.19)
Year FE		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE		Yes	No	Yes	No	Yes	No	Yes	No
Firm FE		No	Yes	No	Yes	No	Yes	No	Yes
Adjusted R^2		0.2187	0.6829	0.2080	0.6914	0.1155	0.6007	0.0763	0.5590
Observations		15,837	15,390	15,837	15,390	15,837	15,390	15,837	15,390

***, **, * Represent significance at the 1 percent, 5 percent, and 10 percent levels, respectively.
This table shows the determinants of AI disclosure. Sample period for this analysis is 2017–2024. Standard errors are clustered by firm. t-statistics are presented in parentheses. Significance is assessed using two-tailed p-values. Under firm fixed effects, 447 observations were dropped due to singleton observations. Results that are consistent between industry and firm fixed effects are highlighted in gray.
Variable definitions are provided in Appendix A.

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effects, aligning with our main findings in Table 4, Panel B. We find an interesting pattern with loss-making firms: although their AI disclosures are not necessarily lengthier, they tend to provide *more* specific AI disclosures, contrary to the idea that firms experiencing difficulties engage in vague "AI washing"-type behavior. These firms could be discussing actual efforts to cut costs or improve operational efficiencies using AI.

Second, our baseline determinants analysis employs OLS, which comes with several limitations: predicted values can fall outside the 0–1 range, and the model assumes constant marginal effects. As a robustness check, we re-estimate the determinants model using conditional logistic regression, as it allows us to control for firm fixed effects. The findings broadly align with our baseline results that larger firms, firms exposed to AI (either through the nature of their business or through the incidence of negative AI-related events), and firms with larger analyst following are more likely to disclose AI, whereas firms undergoing organizational changes are less likely to disclose AI.[15]

Unpacking the Effects of *Ex Ante* and *Ex Post* AI Exposure

To understand our measures of AI exposure better, we first probe the relationship between them. Interestingly, *AIEXPOSURE* and *AIINCIDENT* have very low (and statistically insignificant) correlation; therefore, they capture different aspects of exposure to AI. Although many incidents in the AI Incident database relate to firms/sectors that we would readily associate with AI,[16] incidents also arise in sectors that we would not *prima facie* associate with AI usage.[17] Although *AIEXPOSURE* likely captures specific sectors for which AI creates new business models or products or otherwise fundamentally disrupts the business, *AIINCIDENT* seems to capture a wider range of firms and sectors that are exposed to litigation or reputational consequences through AI-based applications that have been deployed in various functions, often for operational efficiency.

To further probe how AI exposure relates to disclosure choices, we perform a series of (untabulated) additional tests. We start by replacing the *AIEXPOSURE* and *AIINCIDENT* measures in Equation (1) with their one-year lagged versions. Lagged *AIEXPOSURE* has a positive and significant association with *AIDISCLOSE* and *AIBUSINESS*, suggesting that firms with prior AI exposure are more likely to talk about AI. Lagged *AIINCIDENT* is associated with an increased likelihood of disclosing AI risks in Item 1A Risk Factors (*AIRISK*), both in cross-sectional and within-firm tests. Equally, prior AI incidents are unrelated to subsequent AI disclosure in Item 1 Business Description (where we would not necessarily expect an increase). These findings are consistent with reactive transparency, where firms that experience AI-related incidents are more likely to be forthcoming after AI risks materialize. This is resonant with findings in the literature on cybersecurity disclosure, where firms have been documented to step up cybersecurity risk disclosure on several dimensions (length, frequency, and readability) after the advent of cyberattacks (Amani, Magnan, and Moldovan 2025).[18]

Second, we examine whether litigation risk moderates the relationship between AI exposure or incidents and disclosure. Following Francis, Philbrick, and Schipper (1994), we classify certain industries as high litigation risk (*LIT*): biotechnology (SIC 2833–2836, 8731–8734), computers (SIC 3570–3577, 7370–7374), electronics (SIC 3600–3674), and retailing (SIC 5200–5961). We construct two interaction terms, $AIEXPOSURE * LIT$ and $AIINCIDENT * LIT$, and include them in Equation (1). We observe two patterns from this analysis: first, firms in litigious sectors proactively step up AI disclosure when *AIEXPOSURE* is high but more so in Business Description than in Risk Factors. Second, firms experiencing AI incidents increase AI disclosure in all sections, but the $AIINCIDENT * LIT$ interaction is strongly negative. In other words, low-litigation firms step up AI disclosure in the wake of negative AI incidents (reactive transparency), but high-litigation firms display this behavior to a far lesser extent. Taken together, these findings reveal a nuanced disclosure strategy from high-litigation firms: they are more proactive about discussing AI in Business Description when exposure is high but relatively cautious after they experience incidents.

Third, we explore whether high AI exposure firms differ systematically in the focus of their AI disclosure by comparing risk-oriented (Item 1A) and strategy-oriented (Item 7) discussions. We partition firms at the median value of

[15] Findings discussed in this section are tabulated in the Online Appendix. Specifically, the alternative disclosure measures, length of AI discussion, AI topic count, and specificity are reported in Tables OA.1–OA.3 and the conditional logit robustness check is reported in Table OA.4.

[16] For example, litigation over video deepfakes created using popular AI models or accidents caused by self-driving cars.

[17] For example, fast-food chains whose drive-thru voice-ordering systems collected biometric data without customers' consent, retail chains whose in-store video monitors employ facial recognition technology that was faulty, or a range of employers who used AI tools to screen job applicants in a manner that could be potentially discriminatory.

[18] We explore another specification where we augment our baseline Equation (1) with lagged versions of *AIEXPOSURE* and *AIINCIDENT* to include *AIEXPOSURE*, lagged *AIEXPOSURE*, *AIINCIDENT*, and lagged *AIINCIDENT*. *AIEXPOSURE* subsumes lagged *AIEXPOSURE* (consistent with *AIEXPOSURE* reflecting fast-paced evolution in current technologies), but lagged *AIINCIDENT* subsumes *AIINCIDENT* (consistent with reactive transparency—firms beef up AI disclosure after certain events occur that crystallize the negative consequences of AI usage). Consistent with this interpretation, lagged *AIINCIDENT* is again most significant in models of Item 1A Risk Factors and in within-firm specifications; once firms experience negative consequences from AI usage, they step up discussion in Risk Factors.

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AIEXPOSURE into high and low exposure groups. Among low AI exposure firms, we find significantly greater emphasis on AI in Item 1A Risk Factors compared with Item 7 MD&A (mean difference = 0.0084, $p < 0.01$), indicating a risk-focused disclosure strategy. High AI exposure firms provide greater mean levels of both types of discussions, but the gap is smaller (mean difference = 0.0056, $p < 0.10$). Therefore, high AI exposure firms tend to be more balanced in discussing risks versus opportunities from AI, consistent with AI technologies bringing both costs and benefits to these firms. Firms whose business models have not seen significant impacts (yet) from AI are, on the other hand, more one-sided in discussing risks; this aligns with our descriptive findings that more of the discussion in Risk Factors reflects intentions to adopt, whereas MD&A discussion is dominated by firms that have already adopted AI technologies.

Perceptions of AI Disclosure in Form 10-K

As with the determinants test documented in Section V "Determinants of AI Disclosure," we examine how AI disclosure (*AIDISCLOSE*) is related to capital market uncertainty overall, as well as separately by the sections in which AI discussions appear (*AIBUSINESS*, *AIRISK*, and *AIMDA*). Table 5, Panel A presents the descriptive statistics of the sample used for the test of perceptions of AI disclosure. For this analysis, we use a sample period from 2021 to 2023, leaving us with roughly one year before and after the launch of ChatGPT to better capture the differential market perception to AI disclosure pre- and post-ChatGPT. We interpret our results through the lens of voluntary disclosure and information asymmetry frameworks, where disclosures may either reduce or increase uncertainty depending on their informativeness and complexity.

Table 5, Panel B presents the results of estimating Equation (2). In columns (1), (3), and (5), we find that the coefficients on *AIDISCLOSE* are insignificant, consistent with AI disclosure having no discernible impact on the information environment prior to the launch of ChatGPT. Turning to the interaction term, in column (1), we observe that the coefficient on *AIDISCLOSE* * *POSTCHATGPT* is −0.0027 with a p-value < 0.01. In other words, after the launch of ChatGPT in 2022, AI disclosure associates with reduced uncertainty. Consistent with voluntary disclosure and information asymmetry frameworks, this shift implies a higher level of investor attention due to ChatGPT's demonstration of the broader applicability of AI. The breakdown in column (2) shows that this effect is driven by AI disclosures in

TABLE 5

Consequences of AI Disclosure

Panel A: Summary Statistics of the Variables Used in the Consequence Model

Variable	Obs.	Mean	Std. Dev.	Min	Max
POSTFILINGVOL	6,876	0.0296	0.0204	0.0053	0.1823
DISPERSION	5,052	0.0353	0.1597	0.0001	4.5549
UNEXPEARN	5,047	0.0488	0.2225	0.0000	11.5542
*AIDISCLOSE * POSTCHATGPT*	6,876	0.0919	0.2889	0.0000	1.0000
*AIBUSINESS * POSTCHATGPT*	6,876	0.0756	0.2644	0.0000	1.0000
*AIMDA * POSTCHATGPT*	6,876	0.0265	0.1605	0.0000	1.0000
*AIRISK * POSTCHATGPT*	6,876	0.0387	0.1929	0.0000	1.0000
AIDISCLOSE	6,876	0.2309	0.4215	0.0000	1.0000
AIBUSINESS	6,876	0.1944	0.3958	0.0000	1.0000
AIMDA	6,876	0.0695	0.2544	0.0000	1.0000
AIRISK	6,876	0.0848	0.2786	0.0000	1.0000
POSTCHATGPT	6,876	0.3487	0.4766	0.0000	1.0000
BTM	6,876	0.4987	0.8358	−11.5196	13.1516
LNMVE	6,876	7.0587	2.2798	0.9175	13.2464
ABSCAR	6,876	0.0544	0.0637	0.0001	0.7910
PREFILINGVOL	6,876	0.0366	0.0234	0.0089	0.2212
ALPHA	6,876	0.0002	0.0026	−0.0090	0.0265
ANALYSTS	6,876	1.7196	1.1317	0.0000	4.0073
AIEXPOSURE	6,876	0.1744	1.0461	−0.9657	3.2100

This table shows the summary statistics for the consequences of AI disclosure analysis. Sample period for this analysis is 2021–2023. Variable definitions are provided in Appendix A.

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TABLE 5 (continued)

Panel B: AI Disclosure and Information Uncertainty in Capital Markets

	(1)	(2)	(3)	(4)	(5)	(6)
	POSTFILINGVOL		*DISPERSION*		*UNEXPEARN*	
	Coeff. (t-stats)					
AIDISCLOSE	0.0020		−0.0032		−0.0043	
	(1.54)		(−0.58)		(−0.57)	
AIBUSINESS		0.0013		−0.0037		−0.0106
		(0.93)		(−0.60)		(−1.38)
AIRISK		0.0027*		0.0099		0.0096
		(1.70)		(0.97)		(0.88)
AIMDA		0.0002		−0.0094*		0.0030
		(0.13)		(−1.68)		(0.28)
POSTCHATGPT	−0.0021	−0.0022	−0.0366*	−0.0377*	−0.0210	−0.0225
	(−0.76)	(−0.79)	(−1.68)	(−1.74)	(−1.28)	(−1.37)
*AIDISCLOSE * POSTCHATGPT*	−0.0027***		0.0034		0.0031	
	(−3.24)		(0.54)		(0.47)	
*AIBUSINESS * POSTCHATGPT*		−0.0031***		0.0102		0.0133*
		(−3.10)		(1.54)		(1.80)
*AIRISK * POSTCHATGPT*		−0.0018		−0.0087		−0.0123
		(−1.46)		(−0.59)		(−0.80)
*AIMDA * POSTCHATGPT*		0.0015		0.0046		0.0028
		(0.88)		(0.79)		(0.38)
AIEXPOSURE	0.0004	0.0004	0.0044	0.0040	0.0135***	0.0133***
	(0.62)	(0.59)	(1.27)	(1.18)	(2.96)	(2.89)
BTM	−0.0011	−0.0011	0.0683*	0.0683*	0.0553*	0.0553*
	(−1.37)	(−1.39)	(1.76)	(1.76)	(1.86)	(1.86)
LNMVE	−0.0059***	−0.0060***	−0.0041	−0.0040	−0.0109	−0.0106
	(−7.55)	(−7.57)	(−0.34)	(−0.33)	(−1.06)	(−1.02)
ABSCAR	0.0287***	0.0288***	0.0058	0.0066	0.1048*	0.1061*
	(5.53)	(5.56)	(0.09)	(0.10)	(1.88)	(1.88)
PREFILINGVOL	0.1050***	0.1043***	1.7178***	1.7130***	1.9564***	1.9497***
	(3.39)	(3.36)	(3.55)	(3.55)	(3.94)	(3.94)
ALPHA	−0.1621	−0.1593	4.6067***	4.6267***	5.9650***	5.9773***
	(−1.12)	(−1.10)	(3.57)	(3.58)	(4.50)	(4.51)
ANALYSTS			−0.0140	−0.0138	−0.0166	−0.0166
			(−0.71)	(−0.70)	(−0.92)	(−0.93)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.5399	0.5400	0.2862	0.2858	0.3623	0.3620
Observations	6,876	6,876	5,052	5,052	5,047	5,047

***, **, * Represent significance at the 1 percent, 5 percent, and 10 percent levels, respectively.
This table shows the association between AI disclosure and information uncertainty in the capital markets. Sample period for this analysis is 2021–2023. Standard errors are clustered by firm. t-statistics are presented in parentheses. Significance is assessed using two-tailed p-values. Variable definitions are provided in Appendix A.

(continued on next page)

Business Description (*AIBUSINESS*), further indicating that investors responded positively to information about firms' AI applications following ChatGPT's launch.

In columns (3) and (5), we observe no significant differential perception from analysts to AI disclosures before versus after ChatGPT, indicating that AI disclosures overall did not markedly influence analysts' earnings forecasts. However, column (6) reveals that, when firms discuss AI in Business Description in the post-ChatGPT period, analyst

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TABLE 5 (continued)

Panel C: AI Disclosure and Capital Market Information Uncertainty—Subsample with High AI Exposure

	(1)	(2)	(3)	(4)	(5)	(6)
	POSTFILINGVOL		*DISPERSION*		*UNEXPEARN*	
	Coeff. (t-stats)					
AIDISCLOSE	0.0020		−0.0091		−0.0055	
	(1.29)		(−1.38)		(−0.50)	
AIBUSINESS		0.0014		−0.0084		−0.0142
		(0.83)		(−1.13)		(−1.39)
AIRISK		0.0028		−0.0007		0.0046
		(1.35)		(−0.14)		(0.61)
AIMDA		−0.0011		−0.0008		0.0117
		(−0.64)		(−0.14)		(0.76)
POSTCHATGPT	0.0011	0.0010	−0.0314*	−0.0310*	−0.0377*	−0.0387*
	(0.26)	(0.22)	(−1.66)	(−1.65)	(−1.78)	(−1.85)
*AIDISCLOSE * POSTCHATGPT*	−0.0051***		0.0118*		0.0175**	
	(−4.78)		(1.74)		(2.37)	
*AIBUSINESS * POSTCHATGPT*		−0.0053***		0.0062		0.0155*
		(−4.41)		(0.96)		(1.84)
*AIRISK * POSTCHATGPT*		−0.0012		0.0110*		0.0022
		(−0.76)		(1.72)		(0.26)
*AIMDA * POSTCHATGPT*		0.0007		0.0023		0.0064
		(0.34)		(0.33)		(0.64)
AIEXPOSURE	0.0039***	0.0039***	0.0105	0.0105	0.0142	0.0142
	(3.12)	(3.14)	(1.45)	(1.45)	(1.33)	(1.31)
BTM	−0.0004	−0.0005	0.0510	0.0511	0.0399	0.0400
	(−0.43)	(−0.45)	(0.91)	(0.91)	(0.70)	(0.70)
LNMVE	−0.0061***	−0.0062***	−0.0075	−0.0077	−0.0094	−0.0092
	(−6.62)	(−6.74)	(−0.44)	(−0.44)	(−0.55)	(−0.54)
ABSCAR	0.0240***	0.0243***	−0.0602	−0.0611	0.0247	0.0251
	(4.26)	(4.32)	(−0.72)	(−0.73)	(0.70)	(0.72)
PREFILINGVOL	0.1283***	0.1294***	1.8215**	1.8207**	2.2634***	2.2582***
	(3.27)	(3.28)	(2.47)	(2.46)	(2.82)	(2.81)
ALPHA	−0.1454	−0.1383	5.0706***	5.0959***	5.0689***	5.0972***
	(−0.74)	(−0.71)	(3.33)	(3.34)	(3.36)	(3.38)
ANALYSTS			0.0085	0.0084	0.0028	0.0028
			(0.43)	(0.42)	(0.22)	(0.22)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.5920	0.5921	0.3445	0.3431	0.4002	0.3992
Observations	3,347	3,347	2,579	2,579	2,577	2,577

***, **, * Represent significance at the 1 percent, 5 percent, and 10 percent levels, respectively.
This table shows the association between AI disclosure and information uncertainty in the capital markets conditional on high level of *ex ante* AI exposure. Sample period for this analysis is 2021–2023. Standard errors are clustered by firm. t-statistics are presented in parentheses. Significance is assessed using two-tailed p-values.
Variable definitions are provided in Appendix A.

(continued on next page)

forecast errors tend to increase, as indicated by the positive and marginally significant coefficient on *AIBUSINESS * POSTCHATGPT*. Consistent with the notion that more disclosure can at times increase divergence amongst market participants, although AI disclosures reduce investors' uncertainty, analysts seem to face challenges accurately incorporating this information into their forecasts, potentially due to uncertainty in evaluating firms' AI initiatives in the new post-ChatGPT environment. Related to and consistent with this observation, *AIEXPOSURE*, our firm-level measure

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TABLE 5 (continued)

Panel D: AI Disclosure and Capital Market Information Uncertainty—Subsample with Low AI Exposure

	(1)	(2)	(3)	(4)	(5)	(6)
	POSTFILINGVOL		*DISPERSION*		*UNEXPEARN*	
	Coeff. (t-stats)					
AIDISCLOSE	0.0032		0.0072		0.0040	
	(1.60)		(0.59)		(0.32)	
AIBUSINESS		0.0017		−0.0004		−0.0055
		(0.90)		(−0.03)		(−0.39)
AIRISK		0.0021		0.0244		0.0212
		(0.73)		(0.91)		(0.77)
AIMDA		0.0031		−0.0194		−0.0129
		(0.91)		(−1.36)		(−1.10)
POSTCHATGPT	−0.0038	−0.0037	−0.0733	−0.0736	−0.0363	−0.0376
	(−1.05)	(−1.03)	(−1.45)	(−1.46)	(−1.25)	(−1.29)
*AIDISCLOSE * POSTCHATGPT*	0.0011		−0.0131		−0.0199	
	(0.71)		(−0.85)		(−1.27)	
*AIBUSINESS * POSTCHATGPT*		0.0017		0.0170		0.0105
		(0.95)		(1.22)		(0.69)
*AIRISK * POSTCHATGPT*		−0.0011		−0.0399		−0.0376
		(−0.62)		(−1.04)		(−0.95)
*AIMDA * POSTCHATGPT*		−0.0005		0.0031		−0.0010
		(−0.13)		(0.33)		(−0.10)
AIEXPOSURE	0.0006	0.0004	−0.0280	−0.0275	0.0113	0.0120
	(0.27)	(0.17)	(−1.17)	(−1.15)	(0.51)	(0.53)
BTM	−0.0019	−0.0019	0.0879	0.0877	0.0775**	0.0774**
	(−1.43)	(−1.44)	(1.36)	(1.36)	(2.05)	(2.05)
LNMVE	−0.0073***	−0.0073***	−0.0019	−0.0019	−0.0136	−0.0135
	(−5.69)	(−5.66)	(−0.10)	(−0.10)	(−0.85)	(−0.83)
ABSCAR	0.0360***	0.0361***	0.1118	0.1162	0.2469*	0.2511*
	(3.97)	(3.98)	(0.86)	(0.88)	(1.70)	(1.69)
PREFILINGVOL	0.0342	0.0347	1.7246**	1.7177**	1.8496***	1.8458***
	(0.81)	(0.82)	(2.52)	(2.52)	(2.79)	(2.79)
ALPHA	−0.1449	−0.1524	4.9903**	4.9948**	8.1589***	8.1673***
	(−0.73)	(−0.77)	(2.46)	(2.44)	(3.25)	(3.23)
ANALYSTS			−0.0631	−0.0622	−0.0563	−0.0557
			(−1.47)	(−1.47)	(−1.31)	(−1.32)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.5265	0.5253	0.2181	0.2175	0.3047	0.3031
Observations	3,028	3,028	2,152	2,152	2,150	2,150

***, **, * Represent significance at the 1 percent, 5 percent, and 10 percent levels, respectively.
This table shows the association between AI disclosure and information uncertainty in the capital markets conditional on low level of *ex ante* AI exposure Sample period for this analysis is 2021–2023. Standard errors are clustered by firm. t-statistics are presented in parentheses. Significance is assessed using two-tailed p-values.
Variable definitions are provided in Appendix A.

of exposure to AI developments, associates strongly with forecast errors; analysts are in general less accurate at forecasting earnings for AI-exposed firms.

For the control variables, the results are comparable to prior research (Loughran and McDonald 2014; Christensen et al. 2024). Abnormal returns (*ABSCAR*) and pre-filing volatility (*PREFILINGVOL*) are positively

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associated with post-filing stock return volatility (*POSTFILINGVOL*), whereas market value (*LNMVE*) is negatively associated with *POSTFILINGVOL*. Book to market ratio (*BTM*), pre-filing volatility (*PREFILINGVOL*), and pre-filing alpha (*ALPHA*) are positively associated with analysts' earnings forecast dispersion and error. Additionally, *ALPHA* is also positively associated with analysts' earnings forecast error.

Perceptions of AI Disclosure Conditioning on AI Exposure

We further explore the association between AI disclosure and the quality of the information environment by conditioning it on firm exposure to AI. Arguably, the relationship between AI disclosure and uncertainty should be more pronounced for firms that are more exposed to AI compared with those that are less exposed. Table 5, Panel C (Panel D), presents the results for the analysis for observations that have *AIEXPOSURE* above (below) median.

Starting with Table 5, Panel D, for observations with low AI exposure, we do not observe any significant coefficients for our variables of interest. In contrast, Table 5, Panel C, for observations with high AI exposure, affirms our observation from Panel B that investors and analysts perceive AI disclosure differently post-ChatGPT period compared with pre-ChatGPT. Specifically, in Table 5, Panel C, column (1), we continue to see a significant negative relationship between AI disclosure post-ChatGPT (*AIDISCLOSE* $*$ *POSTCHATGPT*) and stock return volatility (*POSTFILINGVOL*). Column (2) indicates that this effect is driven by AI disclosures within Business Description. Turning to analyst variables, columns (3) and (5) show a positive and significant relationship between AI disclosure post-ChatGPT and both forecast dispersion (*DISPERSION*) and forecast error (*UNEXPEARN*). Column (4) suggests that the increased forecast dispersion among analysts is driven by AI disclosures in Risk Factors, whereas column (5) indicates that the rise in forecast error is largely due to AI discussions within Business Description.

Together, these findings suggest that investors and analysts incorporate AI disclosures differently for firms with high AI exposure. AI disclosure appears to reduce uncertainty amongst investors but increases forecast dispersion and forecast error among analysts. This inconsistency may stem from the fact that analysts are generally sophisticated users of financial information (Schipper 1991; Plumlee 2003; Ramnath, Rock, and Shane 2008; Weber 2009; Gibbons, Iliev, and Kalodimos 2020), whereas investors may be more susceptible to market hype and less inclined to critically assess the substance of AI disclosures.

Perceptions of AI Disclosure Conditioning on the Characteristics of AI Disclosures

Utilizing the rich qualitative features extracted in the previous section—the frequency of AI keywords, relative length, sentiment, informational changes, specificity, the number of topics in AI disclosures, and current versus planned use of AI—we examine how perceptions of AI disclosures vary based on these qualitative characteristics.

Table 6, Panel A presents the relationship between post-filing volatility (*POSTFILINGVOL*) and various characteristics of AI disclosure. For the post-ChatGPT period, we observe that greater frequency of AI keyword mentions, longer relative length, a more positive tone, more time-series incremental information, greater specificity, a broader range of topics in AI discussions, and discussions of current AI adoption are associated with lower post-filing volatility. Table 6, Panel B (Panel C) examines the association between analysts' earnings forecast dispersion (forecast error) and the characteristics of AI disclosures. In the post-ChatGPT period, we find that higher frequency of AI keyword mentions, longer relative length, greater time-series incremental information, more specific discussions, more topics in AI discussions, discussions of intention of AI adoption, as well as discussions of current adoption are associated with increased forecast dispersion among analysts; most of these characteristics are consistently linked to larger errors in analysts' earnings forecasts.[19] Overall, these results reinforce the takeaways from Table 5: longer, richer, and more specific AI discussions seem to lower investor uncertainty while heightening analyst uncertainty and disagreement.

VI. CONCLUDING REMARKS

Our study documents the evolution of AI disclosure in 10-K annual reports, examining its determinants and capital market implications. Although AI mentions in Business Description were present even before ChatGPT, AI discussions in Risk Factors surged post-ChatGPT, becoming lengthier, more specific, and increasingly negative in tone. This shift reflects heightened awareness of AI's risks (i.e., security threats, regulatory and legal concerns, reputational damage, and ethical dilemmas), underscoring growing corporate responsibility in AI governance. Leveraging GPT-4o, we identify diverse AI disclosure themes, with firms highlighting AI's role in product innovation, operational efficiency, and

[19] We also utilized cross-sectional incremental information but do not find it to be significantly associated with any of the outcomes post-ChatGPT. Due to content limitation, we do not tabulate the results here.

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TABLE 6

Consequences of AI Disclosure Conditioning on AI Disclosure Characteristics

Panel A: AI Disclosure and Post-Filing Volatility Conditioning on AI Disclosure Characteristics

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	POSTFILINGVOL							
	Coeff. (t-stats)							
*FREQUENCY * POSTCHATGPT*	−0.0001***							
	(−2.68)							
FREQUENCY	0.0001*							
	(1.80)							
*LENGTH * POSTCHATGPT*		−0.0217**						
		(−2.29)						
LENGTH		0.0320*						
		(1.81)						
*POSITIVE * POSTCHATGPT*			−0.0026**					
			(−2.57)					
POSITIVE			0.0002					
			(0.15)					
*ADDINFO * POSTCHATGPT*				−0.0023**				
				(−2.02)				
ADDINFO				0.0000				
				(0.03)				
*SPECIFICITY * POSTCHATGPT*					−0.0006***			
					(−3.01)			
SPECIFICITY					0.0005*			
					(1.69)			
*TOPICS * POSTCHATGPT*						−0.0007***		
						(−3.06)		
TOPICS						0.0004		
						(1.17)		
*INTENTION * POSTCHATGPT*							0.0004	
							(0.30)	
INTENTION							0.0002	
							(0.13)	
*ACTION * POSTCHATGPT*								−0.0021***
								(−3.10)

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TABLE 6 (continued)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	POSTFILINGVOL							
	Coeff. (t-stats)							
ACTION								0.0014
								(1.64)
POSTCHATGPT	−0.0026	−0.0026	−0.0025	−0.0025	−0.0023	−0.0025	−0.0028	−0.0023
	(−0.93)	(−0.93)	(−0.89)	(−0.91)	(−0.84)	(−0.89)	(−1.01)	(−0.84)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.5393	0.5393	0.5396	0.5394	0.5398	0.5395	0.5389	0.5399
Observations	6,876	6,876	6,876	6,876	6,876	6,876	6,876	6,876

***, **, * Represent significance at the 1 percent, 5 percent, and 10 percent levels, respectively.
This table shows the association between various characteristics of AI disclosure (frequency, length, sentiment, information change, specificity, topics, intention to adopt AI, and actual AI adoption) and post-filing volatility. Sample period for this analysis is 2021–2023. Standard errors are clustered by firm. t-statistics are presented in parentheses. Significance is assessed using two-tailed p-values.
Variable definitions are provided in Appendix A.

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TABLE 6 (continued)

Panel B: AI Disclosure and Analysts' Forecast Dispersion Conditioning on AI Disclosure Characteristics

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	DISPERSION							
	Coeff. (t-stats)							
FREQUENCY ∗ *POSTCHATGPT*	0.0004*							
	(1.96)							
FREQUENCY	−0.0001							
	(−0.44)							
LENGTH ∗ *POSTCHATGPT*		0.1020**						
		(2.29)						
LENGTH		−0.0151						
		(−0.32)						
POSITIVE ∗ *POSTCHATGPT*			0.0022					
			(0.27)					
POSITIVE			−0.0022					
			(−0.32)					
ADDINFO ∗ *POSTCHATGPT*				0.0122**				
				(2.21)				
ADDINFO				−0.0087**				
				(−2.14)				
SPECIFICITY ∗ *POSTCHATGPT*					0.0020*			
					(1.92)			
SPECIFICITY					−0.0003			
					(−0.17)			
TOPICS ∗ *POSTCHATGPT*						0.0023*		
						(1.67)		
TOPICS						−0.0007		
						(−0.47)		
INTENTION ∗ *POSTCHATGPT*							0.0081*	
							(1.72)	
INTENTION							−0.0061	
							(−0.91)	
ACTION ∗ *POSTCHATGPT*								0.0069*
								(1.91)
ACTION								−0.0016
								(−0.50)
POSTCHATGPT	−0.0366*	−0.0365*	−0.0358*	−0.0368*	−0.0374*	−0.0370*	−0.0354*	−0.0373*
	(−1.71)	(−1.71)	(−1.67)	(−1.72)	(−1.74)	(−1.72)	(−1.66)	(−1.73)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.2864	0.2864	0.2861	0.2865	0.2865	0.2864	0.2862	0.2865
Observations	5,052	5,052	5,052	5,052	5,052	5,052	5,052	5,052

***, **, * Represent significance at the 1 percent, 5 percent, and 10 percent levels, respectively.
This table shows the association between various characteristics of AI disclosure (frequency, length, sentiment, information change, specificity, topics, intention to adopt AI, and actual AI adoption) and analysts' forecast dispersion. Note: Sample period for this analysis is 2021-2023. Standard errors are clustered by firm. t-statistics are presented in parentheses. Significance is assessed using two-tailed p-values.
Variable definitions are provided in Appendix A.

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TABLE 6 (continued)

Panel C: AI Disclosure and Analysts' Forecast Error Conditioning on AI Disclosure Characteristics

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	UNEXPEARN							
	Coeff. (t-stats)							
FREQUENCY ∗ *POSTCHATGPT*	0.0003*							
	(1.80)							
FREQUENCY	−0.0002							
	(−0.62)							
LENGTH ∗ *POSTCHATGPT*		0.1067**						
		(2.24)						
LENGTH		−0.0530						
		(−0.83)						
POSITIVE ∗ *POSTCHATGPT*			−0.0005					
			(−0.06)					
POSITIVE			−0.0076					
			(−1.01)					
ADDINFO ∗ *POSTCHATGPT*				0.0145**				
				(2.43)				
ADDINFO				−0.0114*				
				(−1.75)				
SPECIFICITY ∗ *POSTCHATGPT*					0.0019*			
					(1.79)			
SPECIFICITY					−0.0008			
					(−0.43)			
TOPICS ∗ *POSTCHATGPT*						0.0021		
						(1.47)		
TOPICS						−0.0019		
						(−0.96)		
INTENTION ∗ *POSTCHATGPT*							0.0106*	
							(1.71)	
INTENTION							−0.0128	
							(−1.36)	
ACTION ∗ *POSTCHATGPT*								0.0073**
								(2.01)
ACTION								−0.0053
								(−1.13)
POSTCHATGPT	−0.0210	−0.0211	−0.0201	−0.0215	−0.0219	−0.0212	−0.0198	−0.0221
	(−1.30)	(−1.31)	(−1.24)	(−1.33)	(−1.35)	(−1.31)	(−1.23)	(−1.36)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.3624	0.3625	0.3624	0.3626	0.3625	0.3624	0.3625	0.3626
n	5,047	5,047	5,047	5,047	5,047	5,047	5,047	5,047

***, **, * Represent significance at the 1 percent, 5 percent, and 10 percent levels, respectively.
This table shows the association between various characteristics of AI disclosure (frequency, length, sentiment, information change, specificity, topics, intention to adopt AI, and actual AI adoption) and analysts' forecast error. Sample period for this analysis is 2021–2023. Standard errors are clustered by firm. t-statistics are presented in parentheses. Significance is assessed using two-tailed p-values.
Variable definitions are provided in Appendix A.

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R&D acceleration. However, as AI adoption rises, so too does the awareness of its risks, particularly among firms with high AI exposure or prior AI-related incidents, where the technology has caused harm to its users.

Capital market participants respond asymmetrically to AI disclosures. Post-ChatGPT, AI disclosure correlates with lower stock return volatility, suggesting that transparency reduces investor uncertainty. However, it also leads to greater analyst forecast dispersion and errors, akin to environmental, social, and governance (ESG) disclosures, where increased information can amplify disagreement in the absence of standardized evaluation frameworks (Christensen, Serafeim, and Sikochi 2022). AI, like ESG, is complex, rapidly evolving, and inherently opaque—complicating efforts to assess its long-term impact on business and society.

Our study has limitations, many of which point to avenues for further inquiry. As one of the first studies to systematically analyze AI disclosures in SEC filings, our findings are primarily descriptive. Our list of AI disclosure determinants is not comprehensive, and what triggers first-time disclosure of AI topics in particular merits further attention. Many other factors could drive or moderate AI disclosure. Future work can also better separate firms that are gesturing toward AI from those that discuss AI topics substantively. When examining capital market responses, we do not establish causal relationships. Future research examining AI disclosures across other channels—such as earnings calls and social media—and incorporating additional firm characteristics, such as governance structures, media coverage, and ESG profile, will help build a more full picture of AI transparency in corporate communication.

Some of our findings raise intriguing questions for future researchers: although AI poses a fundamental threat to the business models and strategies of many sectors—and is acknowledged as such in the public discourse—we see relatively little discussion of these issues in annual reports. Do managers view potential disruptions from an alternative lens of the opportunities they pose (e.g., as AI models become more sophisticated at identifying and exploiting vulnerabilities in source code, does this pose a threat to established vendors of cybersecurity software—or does it open up vast new swathes of business for these firms)? Or does it reflect the constraints of 10-K disclosures—that they are backward-looking and may take time to catch up with the public discourse?

Our findings also have many practical implications for regulators and standard-setters. For the SEC, our evidence of reactive transparency (firms increasing AI risk disclosure following negative AI incidents) suggests that incident-triggered disclosure mandates, analogous to cybersecurity breach notification rules, could be an effective regulatory intervention for improving transparency about AI-related issues. For both the SEC and the IFRS, our evidence showing the divergence between investor and analyst reactions suggests that the capital markets currently lack a standardized AI disclosure framework, a gap that standardized disclosure requirements could help address.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

During the preparation of this work, the authors used ChatGPT to ensure proper grammar and consistency. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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APPENDIX A

Variable	Definition	Source
AIDISCLOSE	Equals 1 if firm *i* discusses AI in 10-K report of year *t* and 0 otherwise.	
AIBUSINESS	Equals 1 if firm *i* discusses AI in Item 1 Business Description of 10-K report of year *t* and 0 otherwise.	
AIRISK	Equals 1 if firm *i* discusses AI in Item 1A Risk Factors of 10-K report of year *t* and 0 otherwise.	
AIMDA	Equals 1 if firm *i* discusses AI in Item 7 MD&A of 10-K report of year *t* and 0 otherwise.	
FREQUENCY	The sum of AI keywords frequencies in Business Description, Risk Factors, and MD&A. This variable equals 0 for observations without AI disclosure in any of these sections.	
LENGTH	The sum of the relative length of AI discussions in Business Description, Risk Factors, and MD&A. This variable equals 0 for observations without AI disclosure in any of these sections.	
POSITIVE	An indicator variable that equals 1 if the sum of the sentiments of AI discussions in Business Description, Risk Factors, and MD&A is positive and 0 otherwise. This variable equals 0 for observations without AI disclosure in any of these sections.	
ADDINFO	The sum of the time-series incremental information of AI discussions in Business Description, Risk Factors, and MD&A. This variable equals 0 for observations without AI disclosure in any of these sections.	
SPECIFICIY	The sum of the specificity score of AI discussions in Business Description, Risk Factors, and MD&A. This variable equals 0 for observations without AI disclosure in any of these sections.	
TOPICS	The sum of main topics discussed in AI discussions of Business Description, Risk Factors, and MD&A. This variable equals 0 for observations without AI disclosure in any of these sections.	
INTENTION	Equals 3 if the firm discusses intention to adopt AI in Business Description, Risk Factors, and MD&A; 2 if intention to adopt AI is discussed in two of the three sections; 1 if intention to adopt AI is discussed in one of the three sections; and 0 if firms do not discuss intention to adopt AI. This variable equals 0 for observations without AI disclosure in any of these sections.	
ACTION	Equals 3 if the firm discusses current adoption of AI in Business Description, Risk Factors, and MD&A; 2 if current adoption of AI is discussed in two of the three sections; 1 if current adoption of AI is discussed in one of the three sections; and 0 if firms do not discuss current adoption of AI. This variable equals 0 for observations without AI disclosure in any of these sections.	
POSTCHATGPT	Equals 1 if the 10-K report was issued after the launch of ChatGPT on November 30, 2022 and 0 otherwise.	
LIT	An indicator variable that equals 1 if the firm operates in a high litigation risk industry (biotechnology, computers, electronics, or retailing) and 0 otherwise.	Compustat
SIZE	Natural logarithm of total assets.	Compustat
ORGCHG	Composite variable that includes asset growth, sales growth, leverage, and merger and acquisition activity.	Compustat
COMPLEXITY	Composite variable that includes the number of segments, the presence of foreign transactions, and the existence of restructuring activities.	Compustat
LOSS	An indicator variable set to 1 if net income is negative and 0 otherwise.	
BETA	The equity beta over the last fiscal period.	CRSP

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APPENDIX A (continued)

Variable	Definition	Source
UNCERTAINTY	The standard deviation of net income over the previous five years, scaled by sales.	Compustat
NEWEQUITY	An indicator that equals 1 if a firm issues new equity or new debt in the following year and 0 otherwise.	Compustat
BIG4	An indicator variable that equals 1 if the firm is audited by one of the Big 4 auditors and 0 otherwise.	Compustat
ANALYSTS	Natural logarithm of 1 plus the number of analysts following.	I/B/E/S
INSTOWN	The percentage of total institutional shares out of total shares outstanding.	Thomson Reuters Institutional Holdings (S34)
AIEXPOSURE	The similarity score between the firm's Item 1 Business Description and AI patents.	SEC EDGAR and Google Patents
AIINCIDENT	The number of AI-related incidents reported for each firm in a given year in the AI incident database at https://incidentdatabase.ai/	The AI Incident Database
POSTFILINGVOL	The root mean squared error from a market model multiplied by 100. The model is estimated using trading days [6, 28] relative to the 10-K filing date, where a minimum of ten observations are required to be included in the sample, following Bonsall et al. (2017) and Christensen et al. (2024).	CRSP
DISPERSION	The standard deviation of analysts' forecasts from the 10-K filing date to the next earnings announcement date divided by the stock price from before the 10-K filing date multiplied by 100. Firm-years must have at least two analysts' forecasts to be included, following Bonsall et al. (2017) and Christensen et al. (2024).	I/B/E/S and CRSP
UNEXPEARN	The absolute value of analyst accuracy multiplied by 100. Analyst accuracy is defined as the actual earnings minus average expected earnings divided by the stock price for firm-years with at least one analyst forecast. We obtain the actual earnings and mean analyst forecast from the I/B/E/S unadjusted data files. To avoid stale forecasts, we include only forecasts occurring between the 10-K filing date and the next earnings announcement date. For analysts with more than one forecast reported during this time interval, we retain only the forecast closest to the filing date in the sample, following Bonsall et al. (2017) and Christensen et al. (2024).	I/B/E/S and CRSP
ALPHA	The alpha from a market model multiplied by 100. The model is estimated using trading days [−252, −6] relative to the 10-K filing date, where a minimum of 60 observations of daily returns must be available to be included in the sample, following Bonsall et al. (2017) and Christensen et al. (2024).	CRSP
PREFILINGVOL	The root mean squared error from a market model multiplied by 100. The model is estimated using trading days [−252, −6] relative to the 10-K filing date, where a minimum of 60 observations are required to be included in the sample, following Bonsall et al. (2017) and Christensen et al. (2024).	CRSP
ABSCAR	The absolute value of the market-adjusted return over the two-day period starting on the 10-K filing date, following Bonsall et al. (2017) and Christensen et al. (2024).	CRSP
BTM	Book value of equity divided by market value of equity.	Compustat
LNMVE	Natural logarithm of the market value of equity of the firm at the end of the year.	Compustat

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